

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)

GRC102 Information Security Governance WEEK 4 LABORATORY

Linux Security Monitoring and Auditing

Audit and Control-Assurance Report: From Technical Evidence to Governance Assurance

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Course / module	GRC102 Information Security Governance/Module 4: Monitoring and Auditing Security Controls
Role in the scenario	Security Control Assurance Analyst
Laboratory environment	ICDFA Ubuntu Practice Hub (assigned Linux VM, reached through the Academy e-campus portal)
Assigned host	student000302 (Ubuntu 24.04.4 LTS)
Laboratory dates	29-30 September 2026
Submission deadline	02 October 2026, 11:59 PM WAT
Evidence	Screenshots are embedded in each module (Figures 1-31); an index is in Appendix C

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1. Executive Summary

This report sets out what I found in the GRC102 Week 4 practical lab. I worked on my assigned ICDFa Ubuntu Practice Hub machine (host **student000302**, Ubuntu 24.04.4 LTS) on 29 and 30 September 2026. Taking the role of a security control assurance analyst, I set up and queried the Linux audit system, reviewed the log sources available, ran a Lynis security assessment, and turned the results into a governance-focused control-monitoring record.

Overall control posture: I could not give assurance over the detective controls on this host. The audit daemon is installed but not running. My custom audit rules were written but could not be loaded. The usual log sources (the systemd journal, auth.log and syslog) are either missing or empty. In practice, that means privileged activity and authentication events on this machine cannot currently be tied to a user or a time. Lynis, an independent tool, backs this up: it finds no running log daemon and recommends enabling auditd. It also gives the host a **hardening index of 62** from 232 tests, with **3 warnings** and **43 suggestions**, and the package manager shows **91 pending updates** that had not changed between 29 and 30 September.

Top three findings, which need governance attention first:

ID	Finding	Priority	Why it matters
W4-F01	Audit system (auditd) not operating; rules not loaded; no audit log	High	Nothing records who touches /etc/passwd, /etc/shadow or which programmes run. The control exists on paper but is not operating, so it cannot be relied on.
W4-F02	No working log management: no journal, auth.log or syslog; login records empty; no log daemon (Lynis LOGG-2130, LOGG-2138)	High	Failed logins, sudo use and system errors cannot be reviewed or sent to a SIEM, so an incident could not be reconstructed.
W4-F03	Patch backlog: 91 pending updates on 29 and 30 Sep; Lynis warning PKGS-7392 (vulnerable packages)	High	Known vulnerabilities may still be exploitable, and the unchanged count suggests nothing was patched in between.

What is working. The host firewall is active, no legacy insecure services (inetd, telnet server) were found, logrotate is present, and the package-database and security-repository checks passed. Several kernel settings already match the Lynis profile, for example kernel.dmesg_restrict, kernel.randomize_va_space and tcp_syncookies (Figures 23, 24, 25, 26).

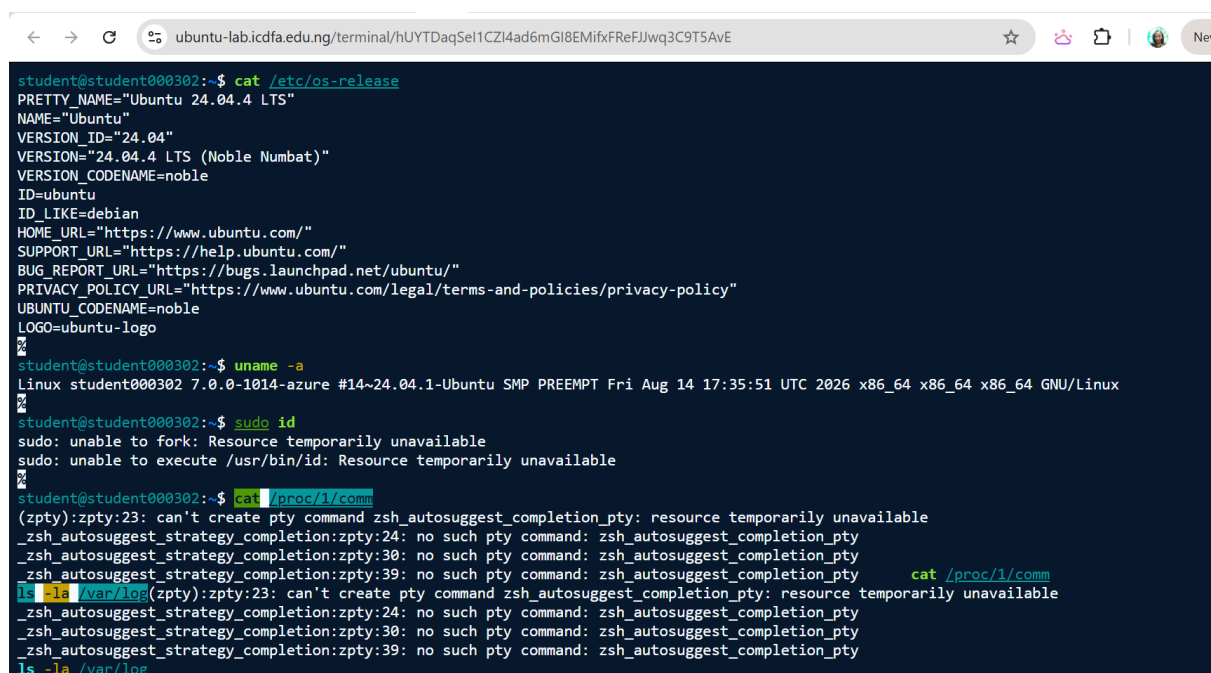
A note on the evidence. PID 1 on this VM is sleep, not systemd, the audit interface refused my rule query (“Operation not permitted”), and the session later hit process limits (“unable to fork”). That points to a restricted, probably containerised, lab platform, and it explains why several standard commands failed. I have kept the real error output as evidence and analysed it as a governance finding. Appendix B maps every requirement in the brief to the evidence I have.

2. Scope and Authorisation

Everything I did was inside the Linux system assigned to me through the ICDFA Ubuntu Practice Hub on the Academy e-campus portal. I did not target any external system, production service or public IP address, and I made no authentication attempts against any other host or account.

Environment	ICDFA Ubuntu Practice Hub: authorised, assigned training VM (browser terminal)
Hostname	student000302
Operating system	Ubuntu 24.04.4 LTS (Noble Numbat), x86_64 (Figure 1)
Kernel	7.0.0-1014-azure #14~24.04.1-Ubuntu SMP PREEMPT, Fri Aug 14 17:35:51 UTC 2026
Account used	student (uid 1000), in the sudo (27) and adm (4) groups among others (Figure 2)
Init system (PID 1)	sleep, so systemd is not the init system (Figure 2)
Laboratory window	29 September 2026 (auditd set-up); 30 September 2026 (auditd queries, log review, Lynis)
Time base	The VM clock is UTC (Figure 3). Screenshot times are from my own device (WAT, which is UTC+1).
Tools in scope	auditd, auditctl, augenrules, ausearch, aureport, journalctl, less, grep, ls, apt, dpkg and Lynis 3.0.9

Environment evidence. The two screenshots below identify the system I worked on: its operating system and kernel, my user and group membership, and the contents of /var/log.

A screenshot of a terminal window with a dark background. The terminal shows the output of several commands: 'cat /etc/os-release' displays Ubuntu 24.04.4 LTS (Noble Numbat) information; 'uname -a' shows the kernel 7.0.0-1014-azure; 'sudo id' fails with 'Resource temporarily unavailable'; 'cat /proc/1/comm' shows 'zsh'; and 'ls -la /var/log' lists log files. The terminal prompt is 'student@student000302:~\$'.

```
student@student000302:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.4 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.4 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo

student@student000302:~$ uname -a
Linux student000302 7.0.0-1014-azure #14~24.04.1-Ubuntu SMP PREEMPT Fri Aug 14 17:35:51 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux

student@student000302:~$ sudo id
sudo: unable to fork: Resource temporarily unavailable
sudo: unable to execute /usr/bin/id: Resource temporarily unavailable

student@student000302:~$ cat /proc/1/comm
(zpty):zpty:23: can't create pty command zsh_autosuggest_completion_pty: resource temporarily unavailable
_zsh_autosuggest_strategy_completion:zpty:24: no such pty command: zsh_autosuggest_completion_pty
_zsh_autosuggest_strategy_completion:zpty:30: no such pty command: zsh_autosuggest_completion_pty
_zsh_autosuggest_strategy_completion:zpty:39: no such pty command: zsh_autosuggest_completion_pty
ls -la /var/log(zpty):zpty:23: can't create pty command zsh_autosuggest_completion_pty: resource temporarily unavailable
_zsh_autosuggest_strategy_completion:zpty:24: no such pty command: zsh_autosuggest_completion_pty
_zsh_autosuggest_strategy_completion:zpty:30: no such pty command: zsh_autosuggest_completion_pty
_zsh_autosuggest_strategy_completion:zpty:39: no such pty command: zsh_autosuggest_completion_pty
ls -la /var/log
```

Figure 1. Operating system, kernel and sudo check

Captured 30 Sep 2026, 21:46 WAT. **Commands:** cat /etc/os-release; uname -a; sudo id. **What it shows:** Ubuntu 24.04.4 LTS (Noble Numbat); kernel 7.0.0-1014-azure on host student000302. sudo id failed with “sudo: unable to fork: Resource temporarily unavailable”.

```

ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
Linux student000302 7.0.0-1014-azure #14~24.04.1-Ubuntu SMP PREEMPT Fri Aug 14 17:35:51 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
uid=1000(student) gid=1000(student) groups=1000(student),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev)
sudo: unable to fork: Resource temporarily unavailable
zsh: segmentation fault (core dumped) sudo id
sleep
total 1028
drwxr-xr-x 1 root root          4096 Sep 30 12:52 .
drwxr-xr-x 1 root root          4096 Jul 30 14:27 ..
lrwxrwxrwx 1 root root           39 Aug 16 03:16 README -> ../../usr/share/doc/systemd/README.logs
-rw-r--r-- 1 root root       17342 Aug 16 04:04 alternatives.log
drwxr-xr-x 1 root root          4096 Sep 30 12:53 apt
drwxr-xr-x 2 root adm           4096 Jul 13 20:10 audit
-rw-r--r-- 1 root root       61229 Jul 30 14:19 bootstrap.log
-rw-rw---- 1 root utmp           0 Jul 30 14:19 bttmp
-rw-r--r-- 1 root root       442785 Sep 30 12:45 dpkg.log
-rw-r--r-- 1 root root           0 Jul 30 14:19 faillog
drwxr-sr-x 2 root systemd-journal 4096 Aug 16 03:16 journal
-rw-rw-r-- 1 root utmp           0 Jul 30 14:19 lastlog
-rw-r----- 1 root root       40281 Sep 30 12:53 lynis-report.dat
-rw-r----- 1 root root      439136 Sep 30 12:53 lynis.log
drwx----- 2 root root          4096 Aug 16 03:16 private
drwxr-xr-x 2 root adm           4096 Jul 24 15:59 samba
-rw-rw-r-- 1 root utmp           0 Jul 30 14:19 wtmp
student@student000302:~$

```

Figure 2. Identity, PID 1 and /var/log inventory

Captured 30 Sep 2026, 21:49 WAT. **Commands:** id; sudo id; cat /proc/1/comm; ls -la /var/log. **What it shows:** uid=1000(student) is a member of the sudo and adm groups. PID 1 is sleep (not systemd). /var/log contains no auth.log, syslog or kern.log; wtmp, bttmp, lastlog and faillog are 0 bytes; audit/ and journal/ directories exist; dpkg.log was last modified at 12:45 (UTC) and lynis.log / lynis-report.dat at 12:53 (UTC).

Changes I made to the system. In line with the lab rules, I did not disable any logging, delete any evidence or change unrelated configuration. What I did change was limited to this:

1. Created /etc/audit/rules.d/custom.rules with five audit rules (Activity 1.2, Figure 8).
2. Ran apt update, which refreshes package lists only (I applied no upgrades), and apt install auditd audispd-plugins, which changed nothing because both were already up to date.
3. Ran systemctl enable auditd, which synchronised the SysV service script (Figure 5).
4. Installed Lynis 3.0.9-1 and its dependency menu 2.1.50 with apt (Figure 16).
5. Opened /etc/passwd with sudo nano and closed it without saving, as the brief instructs (Figure 9).

What I did not do: the optional failed local-login attempt (Activity 1.3) and the optional Activity 3.3 hardening retest.

3. Methodology, Equivalent Sources and Limitations

Area	Tools and commands	Approach
Module 1: auditd	systemctl, service, apt, dpkg, ps, auditctl, augenrules, ausearch, aureport	Checked installation and service state, wrote a dedicated rules file, tried to load and list the rules, generated harmless events, then queried by key and ran summary reports.
Module 2: logs	journalctl, less, grep, ls -la /var/log, id	Reviewed the systemd journal and the traditional authentication and syslog files. Where those were missing, I used the equivalent sources I could find (directory inventory, login-accounting files, Lynis logging checks), as the brief allows.
Module 3: Lynis	lynis --version, lynis audit system	Installed Lynis, ran a full system audit, and recorded the hardening index, warnings, suggestions and report locations.
Governance mapping	Control-monitoring table; finding-record template (Appendix A)	Linked each evidence-backed condition to a control objective, owner, status, threshold, risk, remediation and retest.
Module 4: SIEM	Conceptual mapping table	Showed how host evidence could feed enterprise monitoring, and which conditions are alerts and which are governance issues.

3.1 Where my environment differed from the brief: equivalent sources and methods

The brief asks me to document the equivalent log source or method whenever my distribution or environment differs from what it expects (Section 3 rules, Activities 1.2, 2.2 and 2.3, and the lab access notes). The table gathers every case in one place.

Brief expects	What my environment had	Equivalent source or method I used	Figure
systemctl status / start / restart auditd (Activities 1.1, 1.2)	systemd is not the init system (PID 1 is sleep): "System has not been booted with systemd".	SysV-style sudo service auditd status ("auditd is not running") and ps aux grep -i auditd. systemctl enable completed through systemd-sysv-install.	Figures 5, 6, 7
sudo auditctl -l to record the loaded rules (Activity 1.2)	The audit interface refused the query: "Operation not permitted".	Recorded the rules from the rules file itself (cat /etc/audit/rules.d/custom.rules). A load check with augenrules --check was attempted but blocked by the process limit. I stayed in the assigned environment and recorded what it allowed.	Figures 8, 3
ausearch and aureport records (Activity 1.3)	No /var/log/audit/audit.log exists, and the audit directory is empty.	Documented the exact error output, plus the /var/log/audit and /etc/audit listings, as evidence of the state of the audit subsystem.	Figures 10, 11, 7
journalctl (Activity 2.1)	"No journal files were found" for every variant.	Directory inventory of /var/log and the Lynis logging-daemon checks.	Figures 12, 2, 24
/var/log/auth.log (Activity 2.2)	File does not exist.	Login-accounting files wtmp, btmp, lastlog and faillog (all 0 bytes); id output showing sudo and adm membership; Lynis logging results.	Figures 13, 2, 24
/var/log/syslog (Activity 2.3)	File does not exist.	Lynis warnings and exceptions, and the shell resource errors, as the error and warning evidence. /var/log/apt and dpkg.log exist, but I did not capture their contents.	Figures 14, 21, 4
Auth-log watch "where /var/log/auth.log exists" (Activity 1.2)	auth.log does not exist, yet I added the watch from the brief's example.	Kept the rule as written. On this kind of host it should point at whichever authentication source is enabled instead (see Section 4.2).	Figures 8, 13

Brief expects	What my environment had	Equivalent source or method I used	Figure
Lynis from an approved method (Activity 3.1)	Lynis was not yet installed.	Installed the Ubuntu noble/universe package (Lynis 3.0.9-1) with apt. Lynis itself flags this release as more than four months old.	Figures 16, 17, 27

How I handled the evidence. Every screenshot sits next to the section that discusses it and carries a caption with its capture time, the commands shown and what it demonstrates. The VM clock runs on UTC, and the VM's own file timestamps line up with my screenshot times. For example, dpkg.log was last modified at 12:45 UTC, which matches the Lynis installation screenshot taken at 13:45 WAT.

Limitations.

1. PID 1 is not systemd, so systemctl and journalctl cannot work, and the audit interface refused rule queries ("Operation not permitted").
2. On the evening of 30 September the session hit a process limit ("unable to fork"). That stopped me re-running the auditd status check and augenrules --check (Figure 3, Figure 4).
3. Lynis results describe this lab platform and could differ on a production server. Two Lynis exceptions (PKGS-7410 and KRNL-5788) fit a platform where the kernel is supplied by the host. That is my inference and I have not confirmed it with the platform owner.
4. The KPI/KRI thresholds and remediation timelines in this report are my proposals for management approval. They do not come from the lab brief.

The next two screenshots show the VM clock and the resource errors referred to above.

```

student@student000302:~$ date
Wed Sep 30 20:58:43 UTC 2026
student@student000302:~$ id
uid=1000(student) gid=1000(student) groups=1000(student),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev)
student@student000302:~$ sudo id
sudo: unable to fork: Resource temporarily unavailable
zsh: segmentation fault (core dumped) sudo id
student@student000302:~$ cat /proc/1/comm
sleep
student@student000302:~$ sudo service auditd statussystemctl(zpty):zpty:23: can't create pty command zsh_autosuggest_completion_pty: resource temporarily
unavailable
_zsh_autosuggest_strategy_completion:zpty:24: no such pty command: zsh_autosuggest_completion_pty
_zsh_autosuggest_strategy_completion:zpty:30: no such pty command: zsh_autosuggest_completion_pty
_zsh_autosuggest_strategy_completion:zpty:39: no such pty command: zsh_autosuggest_completion_pty
sudo service auditd statussystemctl
sudo: unable to fork: Resource temporarily unavailable
zsh: segmentation fault (core dumped) sudo service auditd statussystemctl
student@student000302:~$ sudo augenrules --check(zpty):zpty:23: can't create pty command zsh_autosuggest_completion_pty: resource temporarily unavailable
_zsh_autosuggest_strategy_completion:zpty:24: no such pty command: zsh_autosuggest_completion_pty
_zsh_autosuggest_strategy_completion:zpty:30: no such pty command: zsh_autosuggest_completion_pty
_zsh_autosuggest_strategy_completion:zpty:39: no such pty command: zsh_autosuggest_completion_pty
sudo augenrules --check
sudo: unable to fork: Resource temporarily unavailable
zsh: segmentation fault (core dumped) sudo augenrules --check
student@student000302:~$ sudo ls -l /etc/audit/rules.d /var/log/audit(zpty):zpty:23: can't create pty command zsh_autosuggest_completion_pty: resource te
mporarily unavailable
_zsh_autosuggest_strategy_completion:zpty:24: no such pty command: zsh_autosuggest_completion_pty
_zsh_autosuggest_strategy_completion:zpty:30: no such pty command: zsh_autosuggest_completion_pty
_zsh_autosuggest_strategy_completion:zpty:39: no such pty command: zsh_autosuggest_completion_pty

```

Figure 3. System clock, identity, PID 1 and resource errors

Captured 30 Sep 2026, 22:01 WAT. **Commands:** date; id; sudo id; cat /proc/1/comm; sudo service auditd status; sudo augenrules --check. **What it shows:** date reports Wed Sep 30 20:58:43 UTC 2026 (the VM clock is UTC; WAT = UTC+1). sudo commands failed with "unable to fork" and a segmentation fault, so the auditd status re-check and augenrules --check could not execute. Overlapping text is zsh autosuggest output caused by the resource errors.

```
ubuntu-lab.icdfa.edu.ng/terminal/hUYTDaqSel1CZl4ad6mGI8EMifxFReFJJwq3C9T5AvE
student@student000302:~$ bash(zpty):zpty:23: can't create pty command zsh_autosuggest_completion_pty: resource temporarily u
_zsh_autosuggest_strategy_completion:zpty:24: no such pty command: zsh_autosuggest_completion_pty
_zsh_autosuggest_strategy_completion:zpty:30: no such pty command: zsh_autosuggest_completion_pty
_zsh_autosuggest_strategy_completion:zpty:39: no such pty command: zsh_autosuggest_completion_pty

bash: fork: retry: Resource temporarily unavailable
bash: fork: retry: Resource temporarily unavailable
ebash: fork: retry: Resource temporarily unavailable
^M^Xbash: fork: retry: Resource temporarily unavailable
^V^M^C^M^M^M^M^M^Mbash: fork: Resource temporarily unavailable
student@student000302:~$ e
bash: fork: retry: Resource temporarily unavailable
^Mbash: fork: retry: Resource temporarily unavailable
bash: fork: retry: Resource temporarily unavailable
^Mbash: fork: retry: Resource temporarily unavailable
^Cbash: fork: Resource temporarily unavailable
student@student000302:~$
GNU bash, version 5.2.21(1)-release (x86_64-pc-linux-gnu)
student@student000302:~$
student@student000302:~$
student@student000302:~$
student@student000302:~$
student@student000302:~$
student@student000302:~$
student@student000302:~$
student@student000302:~$
student@student000302:~$
```

Figure 4. Shell start-up under process-limit errors

Captured 30 Sep 2026, 22:04 WAT. **Commands:** bash. **What it shows:** GNU bash 5.2.21 started, but repeated “bash: fork: retry: Resource temporarily unavailable” messages show the session had reached a process/resource limit. Stray characters are keystrokes queued while the shell was blocked.

4. Findings - Evidence Bundle 1: auditd Configuration and Events

auditd is the userspace part of the Linux Auditing System. It records security-relevant events such as file access, system calls and user actions, which makes it a source of evidence for forensics, compliance and continuous monitoring. This section covers Activities 1.1 to 1.3: what I did, and what the evidence shows.

4.1 Verifying auditd (Activity 1.1)

Command	What happened	Figure
<code>sudo systemctl status auditd</code>	System has not been booted with systemd as init system (PID 1). Can't operate. Failed to connect to bus: Host is down.	Figure 5
<code>sudo apt install auditd audispd-plugins</code>	Both packages are already the newest version (1:3.1.2-2.1ubuntu0.1); nothing newly installed.	Figures 5, 6
<code>sudo systemctl start auditd</code>	Same systemd error, so the service could not be started this way.	Figures 5, 6
<code>sudo systemctl enable auditd</code>	Synchronised auditd.service with the SysV script through systemd-sysv-install, with no error.	Figure 5
<code>sudo auditctl -s</code>	You must be root to run this program, even with sudo. That is a different message from the later <code>auditctl -l</code> error ("Operation not permitted"). In my reading, both mean the audit control interface is not reachable from this session.	Figures 5, 6
<code>sudo dpkg -l auditd grep auditd</code>	Status ii: auditd 1:3.1.2-2.1ubuntu0.1 amd64 is installed.	Figure 6
<code>cat /proc/1/comm</code>	sleep, so PID 1 is not an init system.	Figures 6, 2
<code>sudo service auditd status</code>	"auditd is not running."	Figure 6
<code>ps aux grep -i auditd</code>	Only the grep process itself appears; there is no auditd process.	Figure 7
<code>sudo ls -l /var/log/audit/</code>	total 0, so the audit log directory is empty.	Figure 7
<code>sudo service auditd status</code> and <code>sudo augenrules --check (re-check, 30 Sep, 22:01)</code>	Neither could run: "sudo: unable to fork: Resource temporarily unavailable".	Figure 3

What this shows: auditd is installed, but it is **not active**. The SysV-style check (`service auditd status`) agrees with the process list.

```
ubuntu-lab.icdfa.edu.ng/terminal/qh0euMB-dig4aNDCK8qcKkOvMPPs3eDHxS-MvTAZx30

student@student000302:~$ sudo systemctl status auditd
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
student@student000302:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
91 packages can be upgraded. Run 'apt list --upgradable' to see them.
student@student000302:~$ sudo apt install auditd audispd-plugins
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
auditd is already the newest version (1:3.1.2-2.1ubuntu0.1).
audispd-plugins is already the newest version (1:3.1.2-2.1ubuntu0.1).
0 upgraded, 0 newly installed, 0 to remove and 91 not upgraded.
student@student000302:~$ sudo systemctl start auditd
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
student@student000302:~$ sudo systemctl enable auditd
Synchronizing state of auditd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable auditd
student@student000302:~$ sudo auditctl -s
You must be root to run this program.
```

Figure 5. auditd status check, package refresh and install attempt (Activity 1.1)

Captured 29 Sep 2026, 13:17 WAT. **Commands:** `sudo systemctl status auditd`; `sudo apt update`; `sudo apt install auditd audispd-plugins`; `sudo systemctl start auditd`; `sudo systemctl enable auditd`; `sudo auditctl -s`. **What it shows:** `systemctl` cannot operate ("System has not been booted with systemd as init system (PID 1)"). `apt update` reports 91 packages can be upgraded. `auditd` and `audispd-plugins` are already the newest version (1:3.1.2-2.1ubuntu0.1). `enable` synchronised with the SysV script; `auditctl -s` returned "You must be root to run this program".

```
ubuntu-lab.icdfa.edu.ng/terminal/qh0euMB-dig4aNDCK8qcKkOvMPPs3eDHxS-MvTAZx30

student@student000302:~$ sudo apt install auditd audispd-plugins
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
auditd is already the newest version (1:3.1.2-2.1ubuntu0.1).
audispd-plugins is already the newest version (1:3.1.2-2.1ubuntu0.1).
0 upgraded, 0 newly installed, 0 to remove and 91 not upgraded.
%
student@student000302:~$ sudo systemctl start auditd
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
%
student@student000302:~$ sudo systemctl enable auditd
Synchronizing state of auditd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable auditd
%
student@student000302:~$ sudo auditctl -s
You must be root to run this program.
%
student@student000302:~$ sudo dpkg -l auditd | grep auditd
ii  auditd      1:3.1.2-2.1ubuntu0.1 amd64      User space tools for security auditing
%
student@student000302:~$ cat /proc/1/comm
sleep
%
student@student000302:~$ sudo service auditd status
* auditd is not running.
%
student@student000302:~$ sudo auditctl -s
You must be root to run this program.
%
student@student000302:~$
```

Figure 6. auditd package and service verification

Captured 29 Sep 2026, 13:28 WAT. **Commands:** `sudo apt install auditd audispd-plugins`; `sudo systemctl start auditd`; `sudo systemctl enable auditd`; `sudo auditctl -s`; `sudo dpkg -l auditd | grep auditd`; `cat /proc/1/comm`; `sudo service auditd status`. **What it shows:** `dpkg` lists `auditd` 1:3.1.2-2.1ubuntu0.1 as installed (ii). PID 1 command is `sleep`. The SysV-compatible service `auditd` status reports “auditd is not running.”

```
ubuntu-lab.icdfa.edu.ng/terminal/EsNpHmvqCHFQ9LheHvIRc6g13sAkxTjgvuW2MQ_1Ai8

%
student@student000302:~$ ps aux | grep -i auditd
student      666  0.0  0.0  3536 1884 pts/0    S+   10:45   0:00 grep --color=auto -i auditd
%
student@student000302:~$ sudo ls -l /var/log/audit/
total 0
%
student@student000302:~$ sudo ls -l /etc/audit/
total 28
-rw-r----- 1 root root  751 Jul 13 20:10 audisp-remote.conf
-rw-r----- 1 root root  127 Jul 13 20:10 audit-stop.rules
-rw-r----- 1 root root  107 Sep 29 07:46 audit.rules
-rw-r----- 1 root root  881 Jul 13 20:10 auditd.conf
drwxr-x---  2 root root 4096 Sep 29 07:46 plugins.d
drwxr-x---  2 root root 4096 Sep 29 12:40 rules.d
-rw-r----- 1 root root  246 Jul 13 20:10 zos-remote.conf
%
student@student000302:~$ sudo cat /etc/audit/auditd.conf | grep log_file
log_file = /var/log/audit/audit.log
max_log_file = 8
max_log_file_action = ROTATE
%
student@student000302:~$
```

Figure 7. auditd process check and audit directory inspection

Captured 30 Sep 2026, 11:50 WAT. **Commands:** `ps aux | grep -i auditd`; `sudo ls -l /var/log/audit/`; `sudo ls -l /etc/audit/`; `sudo cat /etc/audit/auditd.conf | grep log_file`. **What it shows:** No auditd process is running (only the `grep` itself). `/var/log/audit/` is empty (total 0). `/etc/audit/audit.rules` is 107 bytes, modified 29 Sep 07:46 (UTC), earlier than `rules.d` (29 Sep 12:40 UTC). `auditd.conf` sets `log_file=/var/log/audit/audit.log`, `max_log_file=8` and `max_log_file_action=ROTATE`. `audisp-remote.conf` and a `plugins.d` directory are present.

4.2 Writing the audit rules (Activity 1.2)

I created a separate rules file, `/etc/audit/rules.d/custom.rules`, instead of editing the main one. Its contents are in Figure 8 below.

Rule	Key	Purpose and control objective
<code>-w /etc/passwd -p rwx -k passwd_changes</code>	<code>passwd_changes</code>	Record read, write, execute and attribute access to the account database (account integrity).
<code>-w /etc/shadow -p rwx -k shadow_changes</code>	<code>shadow_changes</code>	Record access to the password-hash store (protecting credentials).
<code>-a always,exit -F arch=b64 -S execve -k program_execution</code>	<code>program_execution</code>	Record 64-bit programme execution (who ran what, and when).
<code>-a always,exit -F arch=b32 -S execve -k program_execution</code>	<code>program_execution</code>	The same for 32-bit programmes, so execution cannot slip past monitoring by architecture.
<code>-w /var/log/auth.log -p wa -k auth_failures</code>	<code>auth_failures</code>	Detect writes to, or attribute changes on, the authentication log (a sign of log tampering).

Loading the rules. I could not get them loaded or listed. `sudo systemctl restart auditd` failed because there is no systemd, and `sudo auditctl -l` returned “Error sending rule list data request (Operation not permitted)” followed by “There was an error while processing parameters” (Figure 8). The SysV-style alternatives (`service auditd`, `augenrules --check`) could not start auditd or verify the rules either (Figures 6, 3).

A supporting detail. `/etc/audit/audit.rules` is 107 bytes and was last changed on 29 Sep at 07:46 UTC. That is earlier than the `rules.d` directory, which changed at 12:40 UTC (13:40 WAT) when I created `custom.rules` (Figure 7). It fits with the compiled rule set not having been regenerated to include my file, although I could not run `augenrules --check` to confirm that.

A design point. My `auth_failures` watch points at `/var/log/auth.log`, which does not exist on this host (Figures 13, 2). The brief asks for that watch where the file exists. On a host like this one, the watch should point at whichever authentication log source is enabled instead. The settings in `auditd.conf` (log file `/var/log/audit/audit.log`, 8 MB cap, rotate when full; Figure 7) should also be checked against the organisation’s retention policy.

```

ubuntu-lab.icdfa.edu.ng/terminal/qh0euMB-dig4aNDCK8qcKkOvMPPs3eDHxS-MvTAZx30

student@student000302:~$ sudo nano /etc/audit/rules.d/custom.rules
student@student000302:~$ sudo cat /etc/audit/rules.d/custom.rules
-w /etc/passwd -p rwx -k passwd_changes
-w /etc/shadow -p rwx -k shadow_changes
-a always,exit -F arch=b64 -S execve -k program_execution
-a always,exit -F arch=b32 -S execve -k program_execution
-w /var/log/auth.log -p wa -k auth_failures
student@student000302:~$ sudo systemctl restart auditd
System has not been booted with systemd as init system (PID 1). Can't operate.
Failed to connect to bus: Host is down
student@student000302:~$ sudo auditctl -l
Error sending rule list data request (Operation not permitted)
There was an error while processing parameters
student@student000302:~$

```

Figure 8. Custom audit rules, restart attempt and loaded-rule query (Activity 1.2)

Captured 29 Sep 2026, 13:41 WAT. **Commands:** `sudo nano /etc/audit/rules.d/custom.rules`; `sudo cat /etc/audit/rules.d/custom.rules`; `sudo systemctl restart auditd`; `sudo auditctl -l`. **What it shows:** custom.rules contains the five required rules (passwd_changes, shadow_changes, two program_execution execve rules, auth_failures). systemctl restart fails (no systemd) and auditctl -l returns "Error sending rule list data request (Operation not permitted)", so the loaded rule set could not be displayed.

4.3 Generating and querying audit events (Activity 1.3)

Command	What happened	Figure
<code>sudo nano /etc/passwd</code> (closed without saving)	File opened as root and closed unchanged. No output.	Figure 9
<code>ls /tmp</code>	Listed one entry: icdfa-zsh-1000.	Figure 9
<code>sudo ausearch -k passwd_changes</code>	Error opening /var/log/audit/audit.log (No such file or directory).	Figure 11
<code>sudo ausearch -k program_execution</code>	Error opening /var/log/audit/audit.log (No such file or directory).	Figures 10, 11
<code>sudo ausearch -k auth_failures</code>	Error opening /var/log/audit/audit.log (No such file or directory).	Figure 11
<code>sudo aureport</code>	Summary Report heading, then the same error opening audit.log.	Figures 10, 11
<code>sudo aureport --failed</code>	Failed Summary Report heading, then the same error.	Figure 11
<code>sudo aureport --login</code>	Login Report heading (date, time, auid, host, term, exe, success, event), then the same error.	Figures 10, 11

I ran ausearch for all three keys and all three aureport reports (summary, --failed and --login). Every one returned the same error, on 29 September (Figure 11) and again on 30 September (Figure 10), so the audit log was missing on both days and not just once.

```
ubuntu-lab.icdfa.edu.ng/terminal/EsNpHmvqCHFQ9LheHvIRc6g13sAkxTjgvuW2MQ_1Ai8

student@student000302:~$ sudo nano /etc/passwd
student@student000302:~$ ls /tmp
icdfa-zsh-1000
student@student000302:~$
```

Figure 9. Benign event generation (Activity 1.3)

Captured 30 Sep 2026, 11:24 WAT. **Commands:** `sudo nano /etc/passwd` (exited without saving); `ls /tmp`. **What it shows:** `/etc/passwd` was opened as root and closed without saving; `ls /tmp` listed one entry, `icdfa-zsh-1000`. These are the activities the `passwd_changes` and `program_execution` rules were designed to record.

```
ubuntu-lab.icdfa.edu.ng/terminal/EsNpHmvqCHFQ9LheHvIRc6g13sAkxTjgvuW2MQ_1Ai8

student@student000302:~$ sudo aureport

Summary Report
=====
Error opening /var/log/audit/audit.log (No such file or directory)
student@student000302:~$ sudo ausearch -k program_execution
Error opening /var/log/audit/audit.log (No such file or directory)
student@student000302:~$ sudo aureport --login

Login Report
=====
# date time auid host term exe success event
=====
Error opening /var/log/audit/audit.log (No such file or directory)
student@student000302:~$
```

Figure 10. aureport and ausearch attempts (Activity 1.3)

Captured 30 Sep 2026, 11:26 WAT. **Commands:** `sudo aureport`; `sudo ausearch -k program_execution`; `sudo aureport --login`. **What it shows:** All three tools fail with “Error opening `/var/log/audit/audit.log` (No such file or directory)”. No audit records exist to query or summarise.

```

Error opening /var/log/audit/audit.log (No such file or directory)
student@student000302:~/w4_evidence$ ls /tmp/
icdfa-zsh-1000
student@student000302:~/w4_evidence$ sudo ausearch -k passwd_changes
Error opening /var/log/audit/audit.log (No such file or directory)
student@student000302:~/w4_evidence$ sudo ausearch -k program_execution
Error opening /var/log/audit/audit.log (No such file or directory)
student@student000302:~/w4_evidence$ sudo ausearch -k auth_failures
Error opening /var/log/audit/audit.log (No such file or directory)
student@student000302:~/w4_evidence$ sudo aureport

Summary Report
=====
Error opening /var/log/audit/audit.log (No such file or directory)
student@student000302:~/w4_evidence$ sudo aureport --failed

Failed Summary Report
=====
Error opening /var/log/audit/audit.log (No such file or directory)
student@student000302:~/w4_evidence$ sudo aureport --login

Login Report
=====
# date time auid host term exe success event
=====
Error opening /var/log/audit/audit.log (No such file or directory)
student@student000302:~/w4_evidence$

```

Figure 11. ausearch for all three keys and aureport summaries (Activity 1.3)

Captured 29 Sep 2026, 09:58 WAT. **Commands:** `ls /tmp/`; `sudo ausearch -k passwd_changes`; `sudo ausearch -k program_execution`; `sudo ausearch -k auth_failures`; `sudo aureport`; `sudo aureport --failed`; `sudo aureport --login`. **What it shows:** All three ausearch keys and all three aureport reports (summary, --failed, --login) return “Error opening /var/log/audit/audit.log (No such file or directory)”. The working directory is ~/w4_evidence. The same result appears again on 30 Sep (Figure 10), so the audit log was absent on both days.

4.4 What the evidence says about user, action and time

There is no audit record to read, so I cannot interpret an event directly. The evidence still leads to a clear conclusion. The table compares what an active, correctly loaded rule set would have recorded for my harmless activity in Activity 1.3 with what I could actually see.

Attribute	If auditd were active with the rules loaded	What I could actually see
User	The login UID (auid) and effective UID of the actor. For <code>sudo nano /etc/passwd</code> , this would show user student (uid 1000) acting as root.	Nothing from audit data. Only the shell prompt and <code>id</code> output show the session user is student (Figure 2).
Action	A <code>passwd_changes</code> record for the editor’s access to <code>/etc/passwd</code> , and <code>program_execution</code> records for <code>ls</code> running.	No records. <code>audit.log</code> does not exist, and <code>ausearch</code> and <code>aureport</code> fail (Figures 10, 11).
Time	A kernel-generated timestamp on each record.	Only the screenshot time (30 Sep 2026, about 11:24 WAT) shows when I did it (Figure 9).
Outcome	Success or failure of each operation, plus the executable and key.	Unknown.

Why this matters for governance. The passwd_changes and shadow_changes rules are a detective control over the files that define accounts and credentials. Someone opening /etc/passwd with root rights is exactly the sort of event that has to be attributable, time-stamped and easy to retrieve. With auditd inactive, (1) accountability and non-repudiation for privileged actions cannot be shown, (2) an unauthorised change to account files would go unrecorded, and (3) there is a **gap between how the control was designed and whether it operates**: the rules exist, but the control is not running. Management cannot rely on it, and an internal assurance review would rate it as not operating.

Bundle 1 in summary: I have evidence of the service state (not running), the rules (file contents; loading blocked), and my query and summary attempts (error output), plus the interpretation above. This is finding **W4-F01** (Appendix A).

5. Findings - Evidence Bundle 2: Linux Log Analysis

Linux systems record authentication, service, kernel and application events in systemd journals and in traditional log files. This section covers Activities 2.1 to 2.3, the equivalent sources I used where the standard ones were missing, and a security-event table.

5.1 Reviewing the journal with journalctl (Activity 2.1)

All five commands (`sudo journalctl`, `-u ssh`, `--since "today"`, `-p err` and `-f`) returned "No journal files were found." and "-- No entries --". I stopped the live view (`-f`) with Ctrl+C (Figure 12). A `/var/log/journal` directory does exist (Figure 2), but PID 1 is not systemd, so nothing is running to fill it. **No system-log evidence can be retrieved through the journal.**



```
ubuntu-lab.icdfa.edu.ng/terminal/EsNpHmvqCHFQ9LheHvIRc6g13sAkxTjgyuW2MQ_1Ai8

student@student000302:~$ sudo journalctl
No journal files were found.
-- No entries --
%
student@student000302:~$ sudo journalctl -u ssh
No journal files were found.
-- No entries --
%
student@student000302:~$ sudo journalctl --since "today"
No journal files were found.
-- No entries --
%
student@student000302:~$ sudo journalctl -p err
No journal files were found.
-- No entries --
%
student@student000302:~$ sudo journalctl -f
No journal files were found.
c^C
%
student@student000302:~$
```

Figure 12. journalctl review (Activity 2.1)

Captured 30 Sep 2026, 12:32 WAT. **Commands:** `sudo journalctl`; `sudo journalctl -u ssh`; `sudo journalctl --since "today"`; `sudo journalctl -p err`; `sudo journalctl -f` (stopped with Ctrl+C). **What it shows:** Every variant returns "No journal files were found." / "-- No entries --"; no journal data is available on this host.

5.2 Authentication and privilege use (Activity 2.2)

`sudo less /var/log/auth.log`, `grep -i "failed password"` and `grep -i "sudo"` all returned "No such file or directory" (Figure 13). The brief says to use the equivalent source where a distribution does not use `auth.log`, so I looked at what else this host offers:

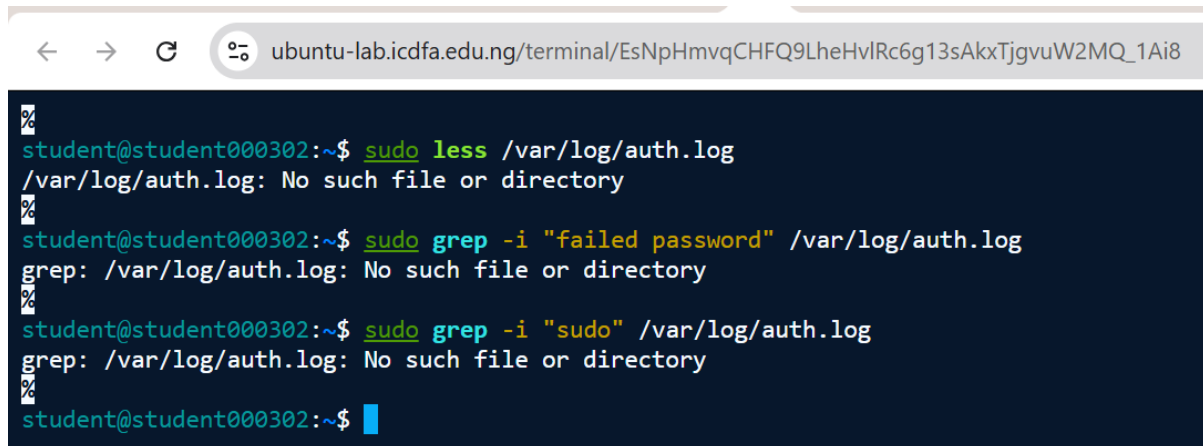
A terminal window with a dark background and light blue text. The window title is 'ubuntu-lab.icdfa.edu.ng/terminal/EsNpHmvqCHFQ9LheHvIRc6g13sAkxTjgvuW2MQ_1Ai8'. The terminal shows three commands being executed: 'sudo less /var/log/auth.log', 'sudo grep -i "failed password" /var/log/auth.log', and 'sudo grep -i "sudo" /var/log/auth.log'. Each command is followed by the output 'grep: /var/log/auth.log: No such file or directory'. The prompt is 'student@student000302:~\$'.

Figure 13. Authentication and privilege-use log review (Activity 2.2)

Captured 30 Sep 2026, 12:39 WAT. **Commands:** `sudo less /var/log/auth.log`; `sudo grep -i "failed password" /var/log/auth.log`; `sudo grep -i "sudo" /var/log/auth.log`. **What it shows:** `/var/log/auth.log` does not exist ("No such file or directory"), so failed-password and sudo events cannot be reviewed from the standard source.

1. **Directory inventory** (Figure 2): `/var/log` has no `auth.log`, `syslog` or `kern.log`. The login-accounting files `wtmp`, `btmpt`, `lastlog` and `faillog` exist but are all **0 bytes** (last modified 30 July), so no successful or failed logins are recorded.
2. **Account privileges** (Figure 2): my user is in the `sudo` group (root through `sudo`) and the `adm` group (read access to log files).
3. **Lynis logging checks** (Figure 24): no log daemon is running, and `klogd` is not running.

Conclusion. I could not observe any significant authentication event. That does not mean none happened. The sources that would record failed passwords and `sudo` use are missing or empty, so a privileged-access review cannot be completed on this host at the moment.

5.3 General system log (Activity 2.3)

`sudo less /var/log/syslog` and the `grep -i "error"` and `grep -i "warning"` searches returned "No such file or directory" (Figure 14). For error and warning evidence from an equivalent source, I used these instead: the three Lynis warnings (Figure 21), the Lynis exceptions (Figure 22) and the "unable to fork" and "fork: retry" errors in my shell (Figures 1, 3, 4). Package-management history (`/var/log/apt` and `dpkg.log`) does exist (Figure 2), but I did not capture its contents.

```

ubuntu-lab.icdfa.edu.ng/terminal/EsNpHmvqCHFQ9LheHvIRc6g13sAkxTjgvuW2MQ_1Ai8

student@student000302:~$ sudo less /var/log/syslog
/var/log/syslog: No such file or directory
%
student@student000302:~$ sudo grep -i "error" /var/log/syslog
grep: /var/log/syslog: No such file or directory
%
student@student000302:~$ sudo grep -i "warning" /var/log/syslog
grep: /var/log/syslog: No such file or directory
%
student@student000302:~$ █

```

Figure 14. General system log review (Activity 2.3)

Captured 30 Sep 2026, 12:41 WAT. **Commands:** `sudo less /var/log/syslog`; `sudo grep -i "error" /var/log/syslog`; `sudo grep -i "warning" /var/log/syslog`. **What it shows:** `/var/log/syslog` does not exist; no syslog error or warning data can be retrieved from the standard source.

5.4 Security-event table

Event / Condition	Evidence Source	Time	User / Service	Security Significance	Recommended Action
E1. journalctl returns no journal files	journalctl (Figure 12)	30 Sep, 12:32 WAT	student / systemd-journald (not running)	No journal evidence is kept, so service and authentication events cannot be reconstructed.	Enable persistent journald or rsyslog and forward to the SIEM (F02).
E2. /var/log/auth.log is missing	auth.log; /var/log listing (Figures 13, 2)	30 Sep, 12:39 and 21:49 WAT	student / sshd, sudo (no log)	Failed logins and privilege use cannot be reviewed.	Log the auth facility, enable sudo logging and review weekly (F02, F05).
E3. /var/log/syslog is missing	syslog (Figure 14)	30 Sep, 12:41 WAT	student / syslog (not present)	System errors and warnings are unavailable.	Install and enable rsyslog to clear LOGG-2130 (F02).
E4. wtmp, btmp, lastlog and faillog are 0 bytes	ls -la /var/log (Figure 2)	30 Sep, 21:49 WAT (files dated 30 Jul)	root:utmp / login accounting	No login or failed-login history is recorded.	Check login accounting works and forward it to the SIEM (F02).
E5. Lynis: no running log daemon; klogd not running	Lynis logging checks (Figures 24, 21)	Scan about 13:53 WAT (12:53 UTC)	lynis (root) / logging subsystem	An independent tool confirms there is no logging; kernel messages may be lost.	Install and enable a log daemon and kernel log capture (F02).
E6. 91 packages upgradable; Lynis PKGS-7392 warning	apt update; Lynis (Figures 5, 15, 21)	29 Sep 13:17; 30 Sep 13:44 WAT	student / apt, dpkg	Pending updates may include security fixes; the count did not change in two days.	Apply updates and enable unattended security upgrades (F03).
E7. Privileged membership (sudo, adm); sudo used during the lab	id output (Figure 2)	30 Sep, 21:49 WAT	student / sudo	Root and log-read rights exist, but their use cannot be independently reviewed.	Record the authorisation, review access quarterly and enable sudo logging (F05).
E8. "unable to fork"; "fork: retry"	Shell output (Figures 1, 3, 4)	30 Sep, 21:46–22:04 WAT	student / shell, sudo	Resource exhaustion blocked commands; security tools could fail under the same limit.	Platform owner to review limits and capacity monitoring (F07).
E9. Lynis exceptions PKGS-7410 and KRNL-5788	Lynis exceptions (Figures 22, 27)	Scan about 13:53 WAT	lynis	Some kernel and package checks cannot be assessed on this platform.	Record as an environment limitation and confirm with the platform owner.

5.5 Routine activity versus real concern

Nothing I collected shows an attack in progress. My own use of sudo to install packages and run Lynis is ordinary lab activity. What **does** need governance attention is the loss of the evidence sources themselves. With no journal, auth log, syslog or login records, the organisation could not detect or investigate brute-force attempts, unauthorised sudo use or service failures on this host. Lynis reports the same thing independently, so this is a verified control weakness and not a quirk of the commands I used. This is finding **W4-F02** (Appendix A).

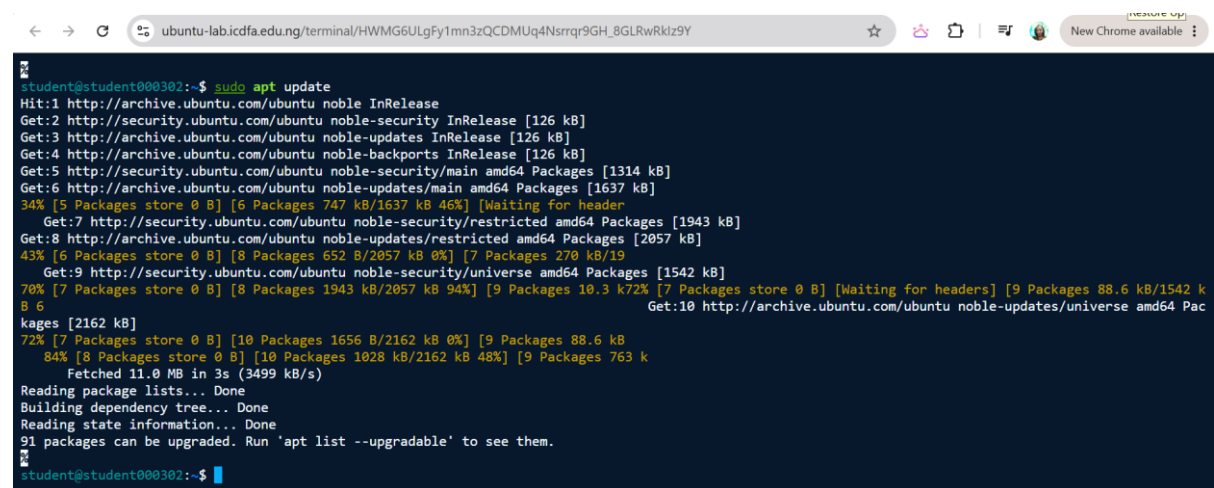
6. Findings - Evidence Bundle 3: Lynis Security Assessment

Lynis is an open-source auditing tool for Unix-like systems. It runs configuration and security checks and reports warnings, suggestions and a hardening index. It assesses a system; it is not a penetration-testing tool.

6.1 Installation, version and scan (Activities 3.1 and 3.2)

I refreshed the package lists, installed Lynis, checked its version and ran a full system audit. The figures below follow that order and end with the scan summary.

Installation	sudo apt install lynis -y installed Lynis 3.0.9-1 and menu 2.1.50 from archive.ubuntu.com (noble/universe) (Figure 16)
Version	Lynis 3.0.9 (Figure 17)
Execution	sudo lynis audit system (Figure 18); the scan finished at about 12:53 UTC (13:53 WAT), going by the log-file modification time (Figure 2)
System detected	Linux, Ubuntu 24.04, kernel 7.0.0, x86_64, hostname student000302
Hardening index	62 (Figure 19)
Tests/plugins	232 tests performed; 1 plugin enabled
Components	Firewall [V]; Malware scanner [X]
Modules	Compliance status [?]; Security audit [V]; Vulnerability scan [V]
Results	3 warnings and 43 suggestions, with an “Exceptions found” notice (Figures 21, 27, 20)
Report and log files	/var/log/lynis-report.dat and /var/log/lynis.log (Figures 18, 19)



```
student@student000302:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1314 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1637 kB]
34% [5 Packages store 0 B] [6 Packages 747 kB/1637 kB 46%] [Waiting for header
Get:7 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1943 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2057 kB]
43% [6 Packages store 0 B] [8 Packages 652 B/2057 kB 0%] [7 Packages 270 kB/19
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1542 kB]
70% [7 Packages store 0 B] [8 Packages 1943 kB/2057 kB 94%] [9 Packages 10.3 kB/72% [7 Packages store 0 B] [Waiting for headers] [9 Packages 88.6 kB/1542 k
8 B 6
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Pac
kages [2162 kB]
72% [7 Packages store 0 B] [10 Packages 1656 B/2162 kB 0%] [9 Packages 88.6 kB
84% [8 Packages store 0 B] [10 Packages 1028 kB/2162 kB 48%] [9 Packages 763 k
Fetched 11.0 MB in 3s (3499 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
91 packages can be upgraded. Run 'apt list --upgradable' to see them.
student@student000302:~$
```

Figure 15. Package list refresh before Lynis installation

Captured 30 Sep 2026, 13:44 WAT. **Commands:** `sudo apt update`. **What it shows:** Ubuntu noble repositories refreshed (11.0 MB fetched). The tool reports “91 packages can be upgraded”, the same count as on 29 Sep (Figure 5).

```
← → ↻ ☰ ubuntu-lab.icdfa.edu.ng/terminal/HWMG6ULgFy1mn3zQCDMUq4Nsrrqr9GH_8GLRwRklz9Y

student@student000302:~$ sudo apt install lynis -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  menu
Suggested packages:
  apt-listbugs debsecan debsums tripwire samhain aide fail2ban menu-l10n gksu
  | kde-cli-tools | ktsuss
The following NEW packages will be installed:
  lynis menu
0 upgraded, 2 newly installed, 0 to remove and 91 not upgraded.
Need to get 602 kB of archives.
After this operation, 3202 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 lynis all 3.0.9-1 [226 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 menu amd64 2.1.50 [377 kB]
Fetched 602 kB in 2s (272 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package lynis.
(Reading database ... 41145 files and directories currently installed.)
Preparing to unpack .../archives/lynis_3.0.9-1_all.deb ...
Unpacking lynis (3.0.9-1) ...
Selecting previously unselected package menu.
Preparing to unpack .../archives/menu_2.1.50_amd64.deb ...
Unpacking menu (2.1.50) ...
Setting up lynis (3.0.9-1) ...
Created symlink /etc/systemd/system/timers.target.wants/lynis.timer → /usr/lib/systemd/system/lynis.timer.
Setting up menu (2.1.50) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for menu (2.1.50) ...
student@student000302:~$
```

Figure 16. Lynis installation (Activity 3.1)

Captured 30 Sep 2026, 13:45 WAT. **Commands:** `sudo apt install lynis -y`. **What it shows:** Lynis 3.0.9-1 and dependency menu 2.1.50 installed from archive.ubuntu.com (noble/universe). apt lists aide, tripwire, samhain, fail2ban and debsums among suggested packages. A lynis.timer symlink was created under systemd timers.

```
← → ↻ ☰ ubuntu-lab.icdfa.edu.ng/terminal/HWMG6ULgFy1mn3zQCDMUq4Nsrrqr9GH_8GLRwRklz9Y

student@student000302:~$ lynis --version
3.0.9
student@student000302:~$
```

Figure 17. Lynis version

Captured 30 Sep 2026, 13:52 WAT. **Commands:** `lynis --version`. **What it shows:** Lynis version 3.0.9.

```
ubuntu-lab.icdfa.edu.ng/terminal/HWMG6ULgFy1mn3zQCDMUq4Nsrrqr9GH_8GLRwRklz9Y

student@student000302:~$ sudo lynis audit system

[ Lynis 3.0.9 ]

#####
Lynis comes with ABSOLUTELY NO WARRANTY. This is free software, and you are
welcome to redistribute it under the terms of the GNU General Public License.
See the LICENSE file for details about using this software.

2007-2021, CISOfy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)
#####

[+] Initializing program
-----
- Detecting OS... [ DONE ]
- Checking profiles... [ DONE ]

-----
Program version:      3.0.9
Operating system:     Linux
Operating system name: Ubuntu
Operating system version: 24.04
Kernel version:       7.0.0
Hardware platform:    x86_64
Hostname:             student000302
-----
Profiles:             /etc/lynis/default.prf
Log file:              /var/log/lynis.log
Report file:          /var/log/lynis-report.dat
Report version:       1.0
```

Figure 18. Lynis audit execution and environment detection (Activity 3.2)

Captured 30 Sep 2026, 13:54 WAT. **Commands:** `sudo lynis audit system`. **What it shows:** Lynis 3.0.9 detects Linux / Ubuntu 24.04, kernel 7.0.0, x86_64, hostname student000302. Profile /etc/lynis/default.prf; log /var/log/lynis.log; report /var/log/lynis-report.dat.

```
ubuntu-lab.icdfa.edu.ng/terminal/HWMG6ULgFy1mn3zQCDMUq4Nsrrqr9GH_8GLRwRkiz9Y

Lynis security scan details:

Hardening index : 62 [##### ]
Tests performed : 232
Plugins enabled : 1

Components:
- Firewall [V]
- Malware scanner [X]

Scan mode:
Normal [V] Forensics [ ] Integration [ ] Pentest [ ]

Lynis modules:
- Compliance status [?]
- Security audit [V]
- Vulnerability scan [V]

Files:
- Test and debug information : /var/log/lynis.log
- Report data : /var/log/lynis-report.dat

=====

Exceptions found
Some exceptional events or information was found!

What to do:
You can help by providing your log file (/var/log/lynis.log).
Go to https://cisofy.com/contact/ and send your file to the e-mail address listed

=====
```

Figure 19. Lynis - scan details and hardening index

Captured 30 Sep 2026, 14:19 WAT. **Commands:** (Lynis output, "Lynis security scan details"). **What it shows:** Hardening index 62; tests performed 232; plugins enabled 1. Components: Firewall [V], Malware scanner [X]. Modules: Compliance status [?], Security audit [V], Vulnerability scan [V]. Files: /var/log/lynis.log and /var/log/lynis-report.dat.

```

ubuntu-lab.icdfa.edu.ng/terminal/HWMG6ULgFy1mn3zQCDMUq4Nsrrqr9GH_8GLRwRklz9Y
- Compliance status      [?]
- Security audit         [V]
- Vulnerability scan     [V]

Files:
- Test and debug information : /var/log/lynis.log
- Report data                : /var/log/lynis-report.dat

=====

Exceptions found
Some exceptional events or information was found!

What to do:
You can help by providing your log file (/var/log/lynis.log).
Go to https://cisofy.com/contact/ and send your file to the e-mail address listed

=====

Lynis 3.0.9

Auditing, system hardening, and compliance for UNIX-based systems
(Linux, macOS, BSD, and others)

2007-2021, CISOfy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)

=====

[TIP]: Enhance Lynis audits by adding your settings to custom.prfl (see /etc/lynis/default.prfl for all settings)

student@student00302:~$

```

Figure 20. Lynis - end of scan and exceptions notice

Captured 30 Sep 2026, 14:20 WAT. **Commands:** (Lynis output, end of run). **What it shows:** Scan completed with the notice “Exceptions found”, report and log file locations, and the closing banner for Lynis 3.0.9.

6.2 Warnings (3)

Lynis ID	Warning	Supporting evidence	Governance significance
PKGS-7392	Found one or more vulnerable packages.	91 pending updates (Figures 5, 15); “Checking vulnerable packages” is WARNING (Figure 22)	The patch and vulnerability-management control is not meeting its objective.
NETW-2705	Couldn’t find 2 responsive nameservers.	Only nameserver 127.0.0.11 is configured (Figure 23)	Single point of failure for name resolution, updates and any log forwarding.
LOGG-2138	klogd is not running, which could lead to missing kernel messages in log files.	klogd NOT FOUND (Figure 24)	Kernel events may never reach the logs; it reinforces W4-F02.

```

-----
=====

-[ Lynis 3.0.9 Results ]-

Warnings (3):
-----
! Found one or more vulnerable packages. [PKGS-7392]
  https://cisofy.com/lynis/controls/PKGS-7392/

! Couldn't find 2 responsive nameservers [NETW-2705]
  https://cisofy.com/lynis/controls/NETW-2705/

! klogd is not running, which could lead to missing kernel messages in log files [LOGG-2138]
  https://cisofy.com/lynis/controls/LOGG-2138/

```

Figure 21. Lynis - Warnings (3)

Captured 30 Sep 2026, 14:16 WAT. **Commands:** (Lynis output, "Warnings (3)"). **What it shows:** PKGS-7392 vulnerable packages found; NETW-2705 could not find 2 responsive nameservers; LOGG-2138 klogd is not running (may lead to missing kernel messages in log files).

6.3 Results by scan section

Section	Key results	Figure
Ports and packages	Vulnerable packages: WARNING. Upgradeable-packages check: SKIPPED. Exception PKGS-7410 (no kernel packages found through the package manager).	Figure 22
Networking	IPv6 enabled; nameserver 127.0.0.11 OK; fewer than two responsive nameservers: WARNING; no promiscuous interfaces.	Figure 23
Firewall	iptables support FOUND; host-based firewall ACTIVE.	Figure 23
Logging and files	Running log daemon check: WARNING. Syslog-NG, systemd journal, Metalog, RSyslog, RFC 3195 daemon and minilogd all NOT FOUND; klogd NOT FOUND. logrotate OK.	Figure 24
Insecure services	inetd and telnet server not found; rsh, NIS and TFTP checks OK.	Figure 24
Kernel hardening	16 sysctl keys reported DIFFERENT from the scan profile (for example fs.protected_fifos, fs.suid_dumpable, kernel.kptr_restrict, kernel.modules_disabled, kernel.sysrq, net.ipv4.conf.all.rp_filter and net.ipv4.conf.all.forwarding). The others were OK.	Figures 25, 26
Hardening checks	Compiler(s) FOUND; malware scanner NOT FOUND.	Figure 26

```
ubuntu-lab.icdfa.edu.ng/terminal/HWMG6ULgFy1mn3zQCDMUq4Nsrrqr9GH_8GLRwRklz9Y

[+] Ports and packages
-----
- Searching package managers
  - Searching dpkg package manager [ FOUND ]
  - Querying package manager
  - Query unpurged packages [ NONE ]
- Checking security repository in sources.list.d directory [ OK ]
- Checking APT package database [ OK ]
- Checking vulnerable packages [ WARNING ]
- Checking upgradeable packages [ SKIPPED ]
- Checking package audit tool [ INSTALLED ]
  Found: apt-get

=====

Exception found!

Function/test: [PKGS-7410]
Message:      Could not find any kernel packages via package manager

Help improving the Lynis community with your feedback!

Steps:
- Ensure you are running the latest version (/usr/sbin/lynis update check)
- If so, create a GitHub issue at https://github.com/CISOfy/lynis
- Include relevant parts of the log file or configuration file

Thanks!

=====
```

Figure 22. Lynis - ports and packages section

Captured 30 Sep 2026, 14:14 WAT. **Commands:** (Lynis output, section "Ports and packages"). **What it shows:** dpkg found; security repository and APT database OK; "Checking vulnerable packages" = WARNING; upgradeable packages check SKIPPED. Exception PKGS-7410: could not find any kernel packages via the package manager.

```
ubuntu-lab.icdfa.edu.ng/terminal/HWMG6ULgFy1mn3zQCDMUq4Nsrrqr9GH_8GLRwRklz9Y

[+] Networking
-----
- Checking IPv6 configuration [ ENABLED ]
  Configuration method [ AUTO ]
  IPv6 only [ NO ]
- Checking configured nameservers
  - Testing nameservers
    Nameserver: 127.0.0.11 [ OK ]
  - Minimal of 2 responsive nameservers [ WARNING ]
- Checking default gateway [ DONE ]
- Getting listening ports (TCP/UDP) [ DONE ]
- Checking promiscuous interfaces [ OK ]
- Checking waiting connections [ OK ]
- Checking status DHCP client
- Checking for ARP monitoring software [ NOT FOUND ]
- Uncommon network protocols [ 0 ]

[+] Printers and Spools
-----
- Checking cups daemon [ NOT FOUND ]
- Checking lp daemon [ NOT RUNNING ]

[+] Software: e-mail and messaging
-----

[+] Software: firewalls
-----
- Checking iptables support [ FOUND ]
- Checking host based firewall [ ACTIVE ]

[+] Software: webserver
```

Figure 23. Lynis - networking and firewall sections

Captured 30 Sep 2026, 14:14 WAT. **Commands:** (Lynis output, sections “Networking” and “Software: firewalls”). **What it shows:** Nameserver 127.0.0.11 OK, but “Minimal of 2 responsive nameservers” = WARNING. No promiscuous interfaces. iptables support FOUND and host-based firewall ACTIVE.

```
ubuntu-lab.icdfa.edu.ng/terminal/HWMG6ULgFy1mn3zQCDMUq4Nsrrqr9GH_8GLRwRklz9Y

[+] Logging and files
-----
- Checking for a running log daemon [ WARNING ]
- Checking Syslog-NG status [ NOT FOUND ]
- Checking systemd journal status [ NOT FOUND ]
- Checking Metalog status [ NOT FOUND ]
- Checking RSyslog status [ NOT FOUND ]
- Checking RFC 3195 daemon status [ NOT FOUND ]
- Checking klogd [ NOT FOUND ]
- Checking minilogd instances [ NOT FOUND ]
- Checking logrotate presence [ OK ]
- Checking log directories (static list) [ DONE ]
- Checking open log files [ SKIPPED ]

[+] Insecure services
-----
- Installed inetd package [ NOT FOUND ]
- Installed xinetd package [ OK ]
  - xinetd status
- Installed rsh client package [ OK ]
- Installed rsh server package [ OK ]
- Installed telnet client package [ OK ]
- Installed telnet server package [ NOT FOUND ]
- Checking NIS client installation [ OK ]
- Checking NIS server installation [ OK ]
- Checking TFTP client installation [ OK ]
- Checking TFTP server installation [ OK ]

[+] Banners and identification
-----
- /etc/issue [ FOUND ]
```

Figure 24. Lynis - logging, insecure services and banners

Captured 30 Sep 2026, 14:14 WAT. **Commands:** (Lynis output, sections "Logging and files", "Insecure services", "Banners and identification"). **What it shows:** "Checking for a running log daemon" = WARNING; Syslog-NG, systemd journal, Metalog, RSyslog, RFC 3195 daemon and minilogd NOT FOUND; klogd NOT FOUND (flagged red). logrotate OK. No inetd or telnet server found; rsh, NIS and TFTP checks OK. /etc/issue FOUND.

[+] Kernel Hardening

```
-----
- Comparing sysctl key pairs with scan profile
- dev.tty.ldisc_autoload (exp: 0) [ OK ]
- fs.protected_fifos (exp: 2) [ DIFFERENT ]
- fs.protected_hardlinks (exp: 1) [ OK ]
- fs.protected_regular (exp: 2) [ OK ]
- fs.protected_symlinks (exp: 1) [ OK ]
- fs.suid_dumpable (exp: 0) [ DIFFERENT ]
- kernel.core_uses_pid (exp: 1) [ DIFFERENT ]
- kernel.ctrl-alt-del (exp: 0) [ OK ]
- kernel.dmesg_restrict (exp: 1) [ OK ]
- kernel.kptr_restrict (exp: 2) [ DIFFERENT ]
- kernel.modules_disabled (exp: 1) [ DIFFERENT ]
- kernel.perf_event_paranoid (exp: 3) [ DIFFERENT ]
- kernel.randomize_va_space (exp: 2) [ OK ]
- kernel.sysrq (exp: 0) [ DIFFERENT ]
- kernel.unprivileged_bpf_disabled (exp: 1) [ DIFFERENT ]
- kernel.yama.ptrace_scope (exp: 1 2 3) [ OK ]
- net.ipv4.conf.all.accept_redirects (exp: 0) [ OK ]
- net.ipv4.conf.all.accept_source_route (exp: 0) [ OK ]
- net.ipv4.conf.all.bootp_relay (exp: 0) [ OK ]
- net.ipv4.conf.all.forwarding (exp: 0) [ DIFFERENT ]
- net.ipv4.conf.all.log_martians (exp: 1) [ DIFFERENT ]
- net.ipv4.conf.all.mc_forwarding (exp: 0) [ OK ]
- net.ipv4.conf.all.proxy_arp (exp: 0) [ OK ]
- net.ipv4.conf.all.rp_filter (exp: 1) [ DIFFERENT ]
- net.ipv4.conf.all.send_redirects (exp: 0) [ DIFFERENT ]
- net.ipv4.conf.default.accept_redirects (exp: 0) [ OK ]
- net.ipv4.conf.default.accept_source_route (exp: 0) [ DIFFERENT ]
- net.ipv4.conf.default.log_martians (exp: 1) [ DIFFERENT ]
- net.ipv4.icmp_echo_ignore_broadcasts (exp: 1) [ OK ]
```

Figure 25. Lynis - kernel hardening (sysctl comparison, part 1)

Captured 30 Sep 2026, 14:15 WAT. **Commands:** (Lynis output, section "Kernel Hardening"). **What it shows:** Comparison of sysctl keys with the scan profile. Several keys are OK (e.g. kernel.dmesg_restrict, kernel.randomize_va_space); others are DIFFERENT (e.g. fs.protected_fifos, fs.suid_dumpable, kernel.kptr_restrict, kernel.modules_disabled, kernel.sysrq, net.ipv4.conf.all.rp_filter).

```

- net.ipv4.conf.default.log_martians (exp: 1) [ DIFFERENT ]
- net.ipv4.icmp_echo_ignore_broadcasts (exp: 1) [ OK ]
- net.ipv4.icmp_ignore_bogus_error_responses (exp: 1) [ OK ]
- net.ipv4.tcp_syncookies (exp: 1) [ OK ]
- net.ipv4.tcp_timestamps (exp: 0 1) [ OK ]
- net.ipv6.conf.all.accept_redirects (exp: 0) [ DIFFERENT ]
- net.ipv6.conf.all.accept_source_route (exp: 0) [ OK ]
- net.ipv6.conf.default.accept_redirects (exp: 0) [ DIFFERENT ]
- net.ipv6.conf.default.accept_source_route (exp: 0) [ OK ]

[+] Hardening
-----
- Installed compiler(s) [ FOUND ]
- Installed malware scanner [ NOT FOUND ]
- Non-native binary formats [ NOT FOUND ]

[+] Custom tests
-----
- Running custom tests... [ NONE ]

[+] Plugins (phase 2)
-----

=====

-[ Lynis 3.0.9 Results ]-

Warnings (3):
-----
! Found one or more vulnerable packages. [PKGS-7392]
https://cisofy.com/lynis/controls/PKGS-7392/

```

Figure 26. Lynis - kernel hardening (part 2), hardening checks and start of results

Captured 30 Sep 2026, 14:16 WAT. **Commands:** (Lynis output, sections “Kernel Hardening”, “Hardening”, “Custom tests”, “Lynis 3.0.9 Results”). **What it shows:** Remaining sysctl results (IPv6 redirect keys DIFFERENT; tcp_syncookies OK). Hardening: compiler(s) FOUND; malware scanner NOT FOUND. Results header and first warning (PKGS-7392).

6.4 The 43 suggestions, grouped by theme

Together, Figures 27, 28, 29, 30, 31 capture all 43 suggestions. The table groups them by theme.

Theme	Lynis IDs	Count	Figure
Logging, accounting, audit and integrity	LOGG-2130, ACCT-9622, ACCT-9626, ACCT-9628, FINT-4350	5	Figure 30
Patch and package management	PKGS-7370, PKGS-7392, PKGS-7394, PKGS-7420, DEB-0810, DEB-0811, DEB-0831	7	Figures 27, 29
Authentication and account policy	AUTH-9230, AUTH-9262, AUTH-9284, AUTH-9286 (x2), AUTH-9328, DEB-0280, DEB-0880	8	Figures 27, 28
Kernel and system hardening	KRNL-5788, KRNL-5820, KRNL-6000, HRDN-7222, HRDN-7230, PROC-3612, BOOT-5180, BOOT-5264	8	Figures 27, 28, 31
Network	NETW-2705, NETW-3200 (dccp, sctp, rds, tipc), NAME-4028	6	Figures 29, 30
File systems and removable media	FILE-6310 (/home, /var), FILE-7524, USB-1000, STRG-1846	5	Figures 28, 29, 31
Legal banners	BANN-7126, BANN-7130	2	Figure 30
Tooling and automation	LYNIS (release older than 4 months), TOOL-5002	2	Figures 27, 30
Total		43	

```
Suggestions (43):
-----
* This release is more than 4 months old. Check the website or GitHub to see if there is an update available. [LYNIS]
  https://cisofy.com/lynis/controls/LYNIS/

* Install libpam-tmpdir to set $TMP and $TMPDIR for PAM sessions [DEB-0280]
  https://cisofy.com/lynis/controls/DEB-0280/

* Install apt-listbugs to display a list of critical bugs prior to each APT installation. [DEB-0810]
  https://cisofy.com/lynis/controls/DEB-0810/

* Install apt-listchanges to display any significant changes prior to any upgrade via APT. [DEB-0811]
  https://cisofy.com/lynis/controls/DEB-0811/

* Install needrestart, alternatively to debian-goodies, so that you can run needrestart after upgrades to determine which daemons are using old version
  s of libraries and need restarting. [DEB-0831]
  https://cisofy.com/lynis/controls/DEB-0831/

* Install fail2ban to automatically ban hosts that commit multiple authentication errors. [DEB-0880]
  https://cisofy.com/lynis/controls/DEB-0880/

* Determine runlevel and services at startup [BOOT-5180]
  https://cisofy.com/lynis/controls/BOOT-5180/

* Consider hardening system services [BOOT-5264]
  - Details : Run '/usr/bin/systemd-analyze security SERVICE' for each service
  https://cisofy.com/lynis/controls/BOOT-5264/

* Determine why /vmlinuz or /boot/vmlinuz is missing on this Debian/Ubuntu system. [KRNL-5788]
  - Details : /vmlinuz or /boot/vmlinuz
  https://cisofy.com/lynis/controls/KRNL-5788/
```

Figure 27. Lynis - Suggestions (43), part 1

Captured 30 Sep 2026, 14:17 WAT. **Commands:** (Lynis output, "Suggestions (43)"). **What it shows:** LYNIS (release older than 4 months), DEB-0280, DEB-0810, DEB-0811, DEB-0831, DEB-0880 (fail2ban), BOOT-5180, BOOT-5264, KRNL-5788.

```
- Details : /vmlinuz or /boot/vmlinuz
  https://cisofy.com/lynis/controls/KRNL-5788/

* If not required, consider explicit disabling of core dump in /etc/security/limits.conf file [KRNL-5820]
  https://cisofy.com/lynis/controls/KRNL-5820/

* Check the output of ps for dead or zombie processes [PROC-3612]
  https://cisofy.com/lynis/controls/PROC-3612/

* Configure password hashing rounds in /etc/login.defs [AUTH-9230]
  https://cisofy.com/lynis/controls/AUTH-9230/

* Install a PAM module for password strength testing like pam_cracklib or pam_passwdqc [AUTH-9262]
  https://cisofy.com/lynis/controls/AUTH-9262/

* Look at the locked accounts and consider removing them [AUTH-9284]
  https://cisofy.com/lynis/controls/AUTH-9284/

* Configure minimum password age in /etc/login.defs [AUTH-9286]
  https://cisofy.com/lynis/controls/AUTH-9286/

* Configure maximum password age in /etc/login.defs [AUTH-9286]
  https://cisofy.com/lynis/controls/AUTH-9286/

* Default umask in /etc/login.defs could be more strict like 027 [AUTH-9328]
  https://cisofy.com/lynis/controls/AUTH-9328/

* To decrease the impact of a full /home file system, place /home on a separate partition [FILE-6310]
  https://cisofy.com/lynis/controls/FILE-6310/

* To decrease the impact of a full /var file system, place /var on a separate partition [FILE-6310]
  https://cisofy.com/lynis/controls/FILE-6310/
```

Figure 28. Lynis - Suggestions, part 2

Captured 30 Sep 2026, 14:17 WAT. **Commands:** (Lynis output, suggestions continued). **What it shows:** KRNL-5820, PROC-3612, AUTH-9230, AUTH-9262, AUTH-9284, AUTH-9286 (minimum and maximum password age), AUTH-9328 (umask 027), FILE-6310 (/home, /var on separate partitions).

```
ubuntu-lab.icdfa.edu.ng/terminal/HWMG6ULgFy1mn3zQCDMUq4Nsrrqr9GH_8GLRwRklz9Y

* To decrease the impact of a full /var file system, place /var on a separate partition [FILE-6310]
  https://cisofy.com/lynis/controls/FILE-6310/

* Disable drivers like USB storage when not used, to prevent unauthorized storage or data theft [USB-1000]
  https://cisofy.com/lynis/controls/USB-1000/

* Disable drivers like firewire storage when not used, to prevent unauthorized storage or data theft [STRG-1846]
  https://cisofy.com/lynis/controls/STRG-1846/

* Check DNS configuration for the dns domain name [NAME-4028]
  https://cisofy.com/lynis/controls/NAME-4028/

* Install debsums utility for the verification of packages with known good database. [PKGS-7370]
  https://cisofy.com/lynis/controls/PKGS-7370/

* Update your system with apt-get update, apt-get upgrade, apt-get dist-upgrade and/or unattended-upgrades [PKGS-7392]
  https://cisofy.com/lynis/controls/PKGS-7392/

* Install package apt-show-versions for patch management purposes [PKGS-7394]
  https://cisofy.com/lynis/controls/PKGS-7394/

* Consider using a tool to automatically apply upgrades [PKGS-7420]
  https://cisofy.com/lynis/controls/PKGS-7420/

* Check your resolv.conf file and fill in a backup nameserver if possible [NETW-2705]
  https://cisofy.com/lynis/controls/NETW-2705/

* Determine if protocol 'dccp' is really needed on this system [NETW-3200]
  https://cisofy.com/lynis/controls/NETW-3200/

* Determine if protocol 'sctp' is really needed on this system [NETW-3200]
  https://cisofy.com/lynis/controls/NETW-3200/
```

Figure 29. Lynis - Suggestions, part 3

Captured 30 Sep 2026, 14:18 WAT. **Commands:** (Lynis output, suggestions continued). **What it shows:** USB-1000, STRG-1846, NAME-4028, PKGS-7370 (debsums), PKGS-7392 (update the system), PKGS-7394, PKGS-7420 (automatic upgrades), NETW-2705, NETW-3200 (dccp, sctp).

```
ubuntu-lab.icdfa.edu.ng/terminal/HWMG6ULgFy1mn3zQCDMUq4Nsrrqr9GH_8GLRwRklz9Y

* Determine if protocol 'sctp' is really needed on this system [NETW-3200]
  https://cisofy.com/lynis/controls/NETW-3200/

* Determine if protocol 'rds' is really needed on this system [NETW-3200]
  https://cisofy.com/lynis/controls/NETW-3200/

* Determine if protocol 'tipc' is really needed on this system [NETW-3200]
  https://cisofy.com/lynis/controls/NETW-3200/

* Check if any syslog daemon is running and correctly configured. [LOGG-2130]
  https://cisofy.com/lynis/controls/LOGG-2130/

* Add a legal banner to /etc/issue, to warn unauthorized users [BANN-7126]
  https://cisofy.com/lynis/controls/BANN-7126/

* Add legal banner to /etc/issue.net, to warn unauthorized users [BANN-7130]
  https://cisofy.com/lynis/controls/BANN-7130/

* Enable process accounting [ACCT-9622]
  https://cisofy.com/lynis/controls/ACCT-9622/

* Enable sysstat to collect accounting (no results) [ACCT-9626]
  https://cisofy.com/lynis/controls/ACCT-9626/

* Enable auditd to collect audit information [ACCT-9628]
  https://cisofy.com/lynis/controls/ACCT-9628/

* Install a file integrity tool to monitor changes to critical and sensitive files [FINT-4350]
  https://cisofy.com/lynis/controls/FINT-4350/

* Determine if automation tools are present for system management [TOOL-5002]
  https://cisofy.com/lynis/controls/TOOL-5002/
```

Figure 30. Lynis - Suggestions, part 4

Captured 30 Sep 2026, 14:18 WAT. **Commands:** (Lynis output, suggestions continued). **What it shows:** NETW-3200 (rds, tipc), LOGG-2130 (syslog daemon), BANN-7126 and BANN-7130 (legal banners), ACCT-9622 (process accounting), ACCT-9626 (sysstat), ACCT-9628 (enable auditd), FINT-4350 (file integrity tool), TOOL-5002.

```

https://cisofy.com/lynis/controls/TOOL-5002/

* Consider restricting file permissions [FILE-7524]
- Details : See screen output or log file
- Solution : Use chmod to change file permissions
https://cisofy.com/lynis/controls/FILE-7524/

* One or more sysctl values differ from the scan profile and could be tweaked [KRNL-6000]
- Solution : Change sysctl value or disable test (skip-test=KRNL-6000:<sysctl-key>)
https://cisofy.com/lynis/controls/KRNL-6000/

* Harden compilers like restricting access to root user only [HRDN-7222]
https://cisofy.com/lynis/controls/HRDN-7222/

* Harden the system by installing at least one malware scanner, to perform periodic file system scans [HRDN-7230]
- Solution : Install a tool like rkhunter, chkrootkit, OSSEC
https://cisofy.com/lynis/controls/HRDN-7230/

Follow-up:
-----
- Show details of a test (lynis show details TEST-ID)
- Check the logfile for all details (less /var/log/lynis.log)
- Read security controls texts (https://cisofy.com)
- Use --upload to upload data to central system (Lynis Enterprise users)

=====

Lynis security scan details:

Hardening index : 62 [#####          ]
Tests performed : 232
Plugins enabled : 1

```

Figure 31. Lynis - Suggestions, part 5 and follow-up

Captured 30 Sep 2026, 14:19 WAT. **Commands:** (Lynis output, suggestions end; “Lynis security scan details” begins). **What it shows:** FILE-7524 (file permissions), KRNL-6000 (sysctl values differ from scan profile), HRDN-7222 (restrict compilers), HRDN-7230 (install a malware scanner such as rkhunter, chkrootkit or OSSEC). Follow-up guidance follows.

6.5 Prioritised findings, owners and recommendations

I picked out the most significant Lynis findings. For each, the table gives the risk or control issue, the evidence, the likely owner, what I recommend and the priority.

#	Finding (Lynis ID)	Evidence	Risk / control issue	Likely owner	Recommended remediation	Priority
1	Vulnerable packages (PKGS-7392)	Figures 21, 15	Known vulnerabilities may be exploitable; the backlog did not change in two days.	Infrastructure / Patch Management Lead	Review <code>apt list --upgradable</code> , apply security updates in a change window and enable unattended security upgrades (PKGS-7420).	High
2	No audit or log daemon (ACCT-9628, LOGG-2130, LOGG-2138)	Figures 24, 30, 21	Detective controls are absent; there is no evidence for incident response.	Security Operations Manager / Platform Engineering	Enable <code>auditd</code> ; install <code>rsyslog</code> (or persistent <code>journald</code>) with kernel log capture; forward to the SIEM.	High
3	No malware scanner or file-integrity tool (HRDN-7230, FINT-4350)	Figures 19, 31, 30	Malware, or unauthorised change to critical files, could go unnoticed.	Endpoint Security Lead	Install a scanner such as <code>rkhunter</code> , <code>chkrootkit</code> or <code>OSSEC</code> (as Lynis lists) and a file-integrity tool such as <code>AIDE</code> , <code>Tripwire</code> or <code>Samhain</code> (named as suggested packages in Figure 16).	Moderate
4	Password and account policy; no brute-force throttling (AUTH-9230/9262/9284/9286/9328, DEB-0880)	Figures 27, 28	Weak password controls and no automatic blocking of repeated failures.	IAM Manager / Platform Engineering	Set hashing rounds, a password-quality PAM module, minimum and maximum age and <code>umask 027</code> in <code>login.defs</code> ; deploy <code>fail2ban</code> or equivalent.	Moderate
5	Sysctl values differ from the profile (KRNL-6000; 16 keys)	Figures 25, 26	Kernel and network hardening is weaker than the baseline.	Platform Engineering	Review each key against operational need and apply through <code>/etc/sysctl.d</code> under change control.	Moderate

#	Finding (Lynis ID)	Evidence	Risk / control issue	Likely owner	Recommended remediation	Priority
6	Single DNS resolver (NETW-2705)	Figure 23	Availability risk for updates and log forwarding.	Network Operations	Configure a backup nameserver by the approved method.	Low
7	No legal banner text (BANN-7126, BANN-7130)	Figure 30	A weaker legal notice to unauthorised users.	Security Policy / Platform Engineering	Add approved banner text to /etc/issue and /etc/issue.net.	Low
8	Lynis release older than 4 months (LYNIS)	Figure 27	An outdated assessment tool may miss newer checks.	Security Assurance	Update Lynis through the approved repository and confirm the version.	Low

6.6 Optional hardening and retest (Activity 3.3)

This optional activity was not performed, and no configuration change was made simply to raise the score, which the brief warns against. A sensible low-risk candidate for a future retest is adding approved legal-banner text (BANN-7126 and BANN-7130). The “before” evidence is that Lynis reports the suggestion (Figure 30), and the “after” would be re-running the relevant Lynis check.

6.7 Consolidated findings register (all bundles)

This register brings the findings from Sections 4 to 6 together, with the evidence, the affected control (as numbered in the control-monitoring table in Section 7), the significance, the priority and the recommended action. The full finding records are in Appendix A.

ID	Finding	Evidence	Affected control	Risk / significance	Priority	Recommended action
W4-F01	Audit system not operating; rules not loaded; no audit log	Figures 6, 8, 10, 11	C1 Security audit logging	No attributable record of account-file access or programme execution.	High	Enable auditd, load the rules and forward to the SIEM.
W4-F02	No working log management	Figures 12, 13, 14, 24	C2 Log management and authentication monitoring	Failed logins, sudo use and errors cannot be reviewed.	High	Enable rsyslog or persistent journald and forward the logs.
W4-F03	Patch backlog; vulnerable packages	Figures 5, 15, 21	C3 Patch and vulnerability management	Known vulnerabilities may be exploitable.	High	Apply security updates; enable unattended security upgrades.
W4-F04	Hardening gaps; no malware scanner or file-integrity tool	Figures 19, 25, 30, 31	C4 Hardening baseline; C6 Malware and file integrity	Weaker baseline; tampering could go unnoticed.	Moderate	Work through the prioritised Lynis suggestions under change control.
W4-F05	Privileged access cannot be independently reviewed	Figure 2	C5 Privileged-access oversight	Cannot verify that sudo use was authorised.	Moderate	Document authorisation, enable sudo logging, recertify quarterly.
W4-F06	Single DNS resolver	Figure 23	C7 Network service resilience	Resolver failure would hold up updates and log forwarding.	Low	Add a backup nameserver.
W4-F07	Platform process limit and restricted environment	Figures 3, 4	Platform (supports C1 and C2)	Limits stop commands and could stop security tools.	Low	Platform owner to review limits and capacity monitoring.

7. Control-Monitoring Table and Governance Escalation (Evidence Bundle 4)

This table links the technical evidence to control objectives, owners, observed status, proposed thresholds, risk, remediation and retest. Rows C1 to C7 rest directly on evidence from this lab. Row C8 is derived from that evidence, because it depends on C1 and C2. The framework references are indicative mappings to ISO/IEC 27002:2022, and the KPI/KRI thresholds are my proposals for management approval. Figure numbers refer to the screenshots in Sections 2 to 6.

Control / objective	Evidence	Owner	Observed status	KPI/KRI or threshold (proposed)	Risk / significance	Remediation	Retest / follow-up
C1. Security audit logging (evidence-based) auditd active and rules loaded, so privileged, account-file and execution activity is attributable. ISO 27002 8.15, 8.16.	auditd installed but not running (Figures 6, 7). custom.rules not loaded; auditctl -l: "Operation not permitted" (Figure 8). No audit.log; ausearch and aureport error (Figures 10, 11). Lynis ACCT-9628 (Figure 30).	Security Operations Manager (operation); System Owner (host)	Not operating	KPI: % of in-scope hosts with auditd active and rules loaded; target 100%. KRI: any host with auditd inactive for more than 24 h goes to the CISO.	No attributable record of account-file access or programme execution; the control cannot be assured.	Enable auditd by the approved platform method; load custom.rules and verify with auditctl -l; forward to the SIEM.	Service active; auditctl -l lists 5 rules; ausearch -k passwd_changes and -k program_execution return records after a harmless test. Within 14 days, then monthly.
C2. Log management and authentication monitoring (evidence-based) Authentication, privilege and system events are logged, kept and reviewable. ISO 27002 8.15.	No journal files (Figure 12). auth.log and syslog missing (Figures 13, 14). wtmp, btmp, lastlog and faillog are 0 bytes (Figure 2). Lynis: no log daemon; klogd not running (Figures 24, 21).	Security Operations Manager; Platform Engineering	Not operating	KPI: log sources reporting / expected = 100%. KRI: a log gap over 1 h alerts the SOC; over 24 h goes to the CISO.	Failed logins, sudo use and errors cannot be reviewed; an incident could not be reconstructed.	Install and enable rsyslog or persistent journald with auth and kernel facilities; set up rotation and forwarding.	Generate a harmless sudo event and confirm it in the auth source and the SIEM. Within 14 days, then a monthly log-coverage review.
C3. Patch and vulnerability management (evidence-based) Security updates are applied within policy. ISO 27002 8.8.	91 upgradable packages on 29 Sep and 30 Sep (Figures 5, 15). Lynis PKGS-7392 warning (Figure 21).	Infrastructure / Patch Management Lead	Backlog present	KPI: critical and high patches applied within SLA (proposed 14 / 30 days). KRI: updates pending beyond SLA go to the Risk function.	Known-vulnerable packages could be exploited; the backlog did not move in two days.	Review pending updates, apply security updates in a change window and enable unattended security upgrades.	apt update shows no pending security updates; PKGS-7392 is gone from a new Lynis run. Within 30 days, then monthly.
C4. System hardening baseline (evidence-based) Configuration meets the approved baseline. ISO 27002 8.9.	Lynis hardening index 62; 232 tests (Figure 19). 16 sysctl keys DIFFERENT (Figures 25, 26). 43 suggestions (Figure 27 to Figure 31).	Platform Engineering / Security Architecture	Partially effective	KPI: hardening index of 80 or more (proposed target). KRI: index below 70, or any new warning, goes to the CISO.	A weaker baseline widens the attack surface.	Work through the prioritised suggestions under change control (password policy, sysctl, services, banners).	Re-run Lynis after changes and every quarter; track the index trend and open warnings.
C5. Privileged-access oversight (evidence-based) Privileged use is authorised and reviewable. ISO 27002 8.2.	My account is in the sudo and adm groups (Figure 2). sudo was used for installs and the audit, and there is no auth log to review it (Figure 13).	IAM Manager / System Owner	Access exists; review evidence unavailable	KPI: 100% of privileged accounts recertified each quarter. KRI: privileged use with no log evidence.	It cannot be verified that sudo use was authorised.	Document the authorisation for sudo and adm membership, enable sudo logging (auth log and auditd) and schedule recertification.	Access-review record plus a sample of sudo events in the SIEM. Quarterly.

Control / objective	Evidence	Owner	Observed status	KPI/KRI or threshold (proposed)	Risk / significance	Remediation	Retest / follow-up
C6. Malware protection and file integrity (evidence-based) Malicious or unauthorised file change is detected. ISO 27002 8.7.	Lynis: Malware scanner [X] (Figure 19); "Installed malware scanner NOT FOUND" (Figure 26). HRDN-7230 and FINT-4350 suggestions (Figures 30, 31).	Endpoint Security Lead	Not implemented	KPI: % of hosts with a scanner and a file-integrity tool. KRI: a change to a critical file with no approved ticket.	Malware or tampering could go unnoticed.	Deploy an approved scanner and file-integrity tool, and baseline the critical files.	Scan report and integrity baseline produced. Within 60 days.
C7. Network service resilience (DNS) (evidence-based) Name resolution stays available. ISO 27002 8.20.	Lynis NETW-2705 warning; single nameserver 127.0.0.11 (Figures 23, 21).	Network Operations	Warning	KPI: two or more responsive resolvers per host.	A resolver failure would hold up updates and any log forwarding.	Add a backup nameserver by the approved method.	NETW-2705 no longer reported. Within 30 days.
C8. Centralised monitoring: SIEM forwarding (derived) Host evidence reaches the SIEM for correlation and alerting. ISO 27002 8.16.	I did not assess this directly. It depends on C1 and C2: at the moment there is no working local log or audit source to forward.	Security Operations Manager	Not evidenced	KPI: % of in-scope hosts forwarding audit and auth logs. KRI: a host silent for more than 24 h.	Without forwarding, gaps on the host are invisible to the SOC and to management.	Once C1 and C2 are fixed, onboard the host to the SIEM (see Section 8).	Test events visible in the SIEM within 60 days of C1 and C2 closing.

7.1 Governance escalation questions

Question	Answer
Which finding is the highest priority, and why?	W4-F01, the audit system not operating , together with the closely linked W4-F02 (no log management). Every other control is only as assurable as the evidence behind it. With no audit or log records, unauthorised changes to account files, privileged activity and authentication attacks cannot be detected, attributed or investigated, and management cannot be sure that other controls work. W4-F03 (the patch backlog) comes next because it can be exploited directly.
Who should own remediation for each major finding?	F01 and F02: Security Operations Manager (running the control) with Platform Engineering (host configuration). F03: Infrastructure / Patch Management Lead. F04 (hardening, malware protection, file integrity): Platform Engineering with the Endpoint Security Lead. F05: IAM Manager and System Owner. F06: Network Operations. F07: the platform owner.
What should trigger escalation to the CISO, the Risk function or the management committee? (my proposal)	To the CISO: auditd or log collection inactive on an in-scope host for more than 24 h; any unapproved change to /etc/passwd or /etc/shadow; repeated authentication failures across several accounts or against a privileged account; a critical or high patch beyond its SLA; a Lynis hardening index below 70 or a new warning. To the Risk function: a control failure open for more than 30 days, or one that comes back after a failed retest, recorded in the risk register. To the management committee: a detective control failing on a business-critical workload for more than 60 days, any request to accept the risk, or repeated failed retests.
What evidence would show the remediation worked?	F01: auditd active; auditctl -l lists the five rules; ausearch and aureport return records with user, action and time after a harmless test. F02: log files or the journal populated; a harmless sudo or failed-login test visible locally and in the SIEM. F03: apt update shows no pending security updates, and PKGS-7392 is absent from a new Lynis run. F04 to F07: the related Lynis items cleared and the hardening index at or above target.

Question	Answer
When and how should each control be retested?	By re-running the same commands I used in this lab (status, rule listing, ausearch and aureport, log greps, lynis -audit -system) and comparing the results with the “before” figures in Sections 4 to 6. My proposed schedule is: F01 and F02 within 14 days of remediation, then monthly; F03 within 30 days, then monthly; F04 to F07 at the next quarterly Lynis assessment. In an enterprise setting, the SIEM checks in Section 8 would run these tests continuously.

8. SIEM/Continuous Monitoring Mapping (Evidence Bundle 5)

One-page view of how the evidence from this lab could feed an enterprise SIEM and continuous monitoring, and which conditions stay technical alerts and which become governance issues. Figure numbers point to my screenshots.

1. Linux host evidence	2. Collection and transport	3. SIEM processing	4. SOC (operational alert)	5. Governance / control assurance
auditd, journal / syslog, auth logs, wtmp and btmp, Lynis report, package status	Audit dispatcher plugins (auditd-remote present, Figure 7); rsyslog or an agent over TLS; scheduled Lynis run (lynis.timer, Figure 16)	Normalise and parse (audit, uid, exe, key, time); correlate by user, host and time; enrich with asset owner; retain per policy	Rule or analytic alert, SOC notified, ticket opened, triage and contain	Dashboards, automated control tests and reports; breaches escalated to CISO / Risk; remediation tracked to closure

Evidence source and what the lab showed	How it reaches the SIEM	Correlation and analysis	Stays a technical operational alert	Becomes a governance / control-assurance issue
auditd (passwd_changes, shadow_changes, program_execution). Rules written; service not running; no records (Figures 8, 10, 11).	Forward audit events by dispatcher plugin or agent; parse key=value fields.	Link account-file access and execve by the same audit; flag activity outside change windows.	A write to /etc/shadow, or an unexpected privileged programme running.	auditd inactive or silent for more than 24 h; rules missing against baseline (C1).
Journal / syslog. No journal, no syslog, no log daemon (Figures 12, 14, 24).	Forward through a syslog daemon or agent over TLS.	Normalise severity; spot service crashes and error spikes.	A service failure or burst of repeated errors on a host.	No log daemon on an in-scope host; coverage below target (C2).
Authentication logs, wtmp, btmp. auth.log missing; login files 0 bytes (Figures 13, 2).	Collect the auth facility and login-accounting data.	Correlate failed logins by source and account; flag sudo use out of hours.	A brute-force pattern from one source; a success after several failures.	No authentication source; repeated failures across accounts; privileged use that cannot be reviewed (C2, C5).
Lynis report (/var/log/lynis-report.dat). Index 62; 3 warnings; 43 suggestions (Figures 19, 21).	Ship the report file on a schedule; parse hardening_index and warnings.	Trend the index per host against the baseline.	A new critical warning raises a ticket to the owner.	Index below threshold; warnings open beyond SLA; declining trend (C4).
Package and patch status. 91 pending updates (Figures 5, 15).	Report inventory and update status; integrate a vulnerability scanner.	Map pending updates to known vulnerabilities.	A package with a known, actively exploited vulnerability.	A breach of the patch SLA (C3).
Platform health. "unable to fork" errors (Figures 1, 4).	Host metrics and agent health.	Correlate resource exhaustion with failing security tools.	Process or memory exhaustion on a host.	Recurring capacity limits that stop security tools running (F07).

8.1 Automation and continuous control monitoring

- Continuous control monitoring and automated control testing:** scheduled scripts check that auditd is active, the five rules are loaded, a log daemon is running and the pending-update count is within its threshold, and send the results to the SIEM dashboard.
- Automated alerting and trend analysis:** a breached threshold (for example, a host silent for more than 24 h) notifies the SOC automatically, and the Lynis hardening index and warning counts are tracked for each host over time.
- Automated reporting and workflow:** recurring control reports go to management, and findings create tickets that are assigned, escalated and tracked to closure through a SOAR or ticketing workflow.

Alert or governance issue? A single event, such as a burst of failed logins or an unexpected write to /etc/shadow, is a technical alert for the SOC to triage. A missing or failing control (no auditd, no log source, a patch backlog beyond SLA, a hardening index below threshold) is a governance and control-assurance issue, because it affects whether

the organisation can detect and respond at all. It needs a named owner, a threshold to track against, and escalation to the CISO, the Risk function or the management committee.

9. Remediation and Retest Plan

The owners, priorities and timelines below are my proposals for management approval. Each action lists the evidence that would close it and how it would be retested.

Ref	Remediation action	Owner	Priority	Proposed timeline	Closure evidence	Retest method and timing
F01	Enable auditd by the approved platform method; load custom.rules; add an equivalent authentication-source watch; forward to the SIEM.	Security Operations Manager; Platform Engineering	High	14 days	Service active; auditctl -l shows 5 rules; audit.log present.	ausearch -k ... and aureport after a harmless test; monthly control check.
F02	Install and enable rsyslog or persistent journald with auth and kernel facilities; check login accounting; forward logs.	Security Operations Manager; Platform Engineering	High	14 days	Auth and syslog sources populated; LOGG-2130 and LOGG-2138 cleared.	A harmless sudo and failed-login test visible locally and in the SIEM; monthly coverage review.
F03	Review and apply pending updates; enable unattended security upgrades.	Infrastructure / Patch Management Lead	High	30 days (critical: 14 days)	No pending security updates; PKGS-7392 cleared.	apt update and a new lynis audit system; monthly.
F04	Work through the prioritised Lynis suggestions under change control: password policy, fail2ban, sysctl values, malware scanner, file-integrity tool, banners.	Platform Engineering; Endpoint Security Lead; IAM Manager	Moderate	60 days	Related Lynis IDs cleared; hardening index at or above target.	Quarterly lynis audit system; compare the index trend.
F05	Document authorisation for sudo and adm membership; enable sudo logging; recertify privileged access.	IAM Manager; System Owner	Moderate	30 days	Approved access record; sudo events visible in logs.	Quarterly access review with a log sample.
F06	Add a backup DNS resolver.	Network Operations	Low	30 days	NETW-2705 cleared.	Next Lynis run.
F07	Review process and resource limits for the lab platform and monitor capacity.	Platform owner	Low	14 days	No “unable to fork” errors in normal use.	Repeat the lab command set; capacity dashboard.

10. Conclusion

The lab gave me solid evidence that, on the Linux machine I was assigned, the detective controls an organisation would depend on (system auditing and authentication monitoring) are not operating. auditd is installed but inactive, my rules are staged but not loaded, and the journal, auth.log and syslog sources are missing or empty. Lynis confirms the logging gap independently and adds a patch backlog (91 pending updates and warning PKGS-7392), a hardening index of 62 and 43 suggestions for improvement. The main governance point is that these are failures of control operation, not just configuration details. Until audit and log evidence is restored, management cannot be assured about privileged activity, authentication attacks or changes to sensitive files on this host.

I would restore audit and log evidence first (W4-F01 and W4-F02), then clear the patch backlog (W4-F03), then work through the hardening suggestions under change control. Each control should be retested with the same commands I used here and, in an enterprise setting, monitored continuously through a SIEM with clear escalation thresholds. The limits of the lab platform (no systemd, a restricted audit interface and process limits) are set out in Section 3, and they should be confirmed with the platform owner.

What I encountered, and what it taught me. The lab did not go the way the commands in the brief assume. auditd would not start, the audit interface refused my queries, the usual log files were missing, and my session later ran out of processes. Rather than hide any of that, I recorded it exactly as it appeared and explained what each error meant. Whether the platform is deliberately left this way or is simply restricted, the effect is the same: it showed me that a control can be approved, installed and even configured on paper and still produce no evidence. That is the difference between a control existing and a control operating, which is the point of this module. I can explain every command I ran, what each result or error told me, and how I would take the same gaps to an owner, a threshold and a retest.

11. References

1. ICDFA. GRC102 Week 4 Practical Laboratory: Linux Security Monitoring and Auditing (laboratory brief, 2026).
2. CISOfy. Lynis security auditing tool and control documentation. <https://cisofy.com/lynis/> and <https://cisofy.com/lynis/controls/> (the control pages cited in the scan output).
3. Linux Audit documentation and manual pages: auditd(8), auditctl(8), augenrules(8), ausearch(8), aureport(8).
4. ISO/IEC 27002:2022, Information security controls: 8.2 (Privileged access rights), 8.7 (Protection against malware), 8.8 (Management of technical vulnerabilities), 8.9 (Configuration management), 8.15 (Logging), 8.16 (Monitoring activities), 8.20 (Networks security). Used as an indicative mapping.
5. NIST SP 800-92, Guide to Computer Security Log Management (2006).

Appendix A — Security Event / Finding Records

Each record follows the template in the lab brief. Times are WAT unless stated. The figures are embedded in Sections 2 to 6 and indexed in Appendix C.

W4-F01 — Audit system not operating

Finding ID	W4-F01
Date / time observed	29 Sep 2026, 13:17–13:41; 30 Sep 2026, 11:24–11:50 and 22:01
Evidence source	auditd, auditctl, ausearch, aureport, service status, ps, /etc/audit and /var/log/audit listings; Lynis ACCT-9628
Evidence summary	auditd 1:3.1.2-2.1ubuntu0.1 is installed but “not running”; systemctl is unusable (PID 1 is sleep); custom.rules was created but auditctl -l returns “Operation not permitted”; /var/log/audit is empty; ausearch and aureport fail with “Error opening /var/log/audit/audit.log”. (Figures 5, 6, 8, 10, 11, 7, 3)
Security significance	There is no attributable record of access to /etc/passwd or /etc/shadow, or of programme execution. The control exists on paper but is not operating, so it cannot be assured.
Control objective	Affected control: C1 (security audit logging). Security-relevant events on sensitive files and process execution are recorded, attributable to a user and time, retained and available for investigation.
Control owner	Security Operations Manager (operation); System Owner / Platform Engineering (host configuration)
Risk / priority	[object Object]
Recommended action	Enable auditd by the approved platform method; load custom.rules (augenrules --load) and verify with auditctl -l; use an equivalent authentication-source watch where auth.log does not exist; forward events to the SIEM; confirm retention (auditd.conf: 8 MB, ROTATE).
Retest evidence	Status active; auditctl -l lists 5 rules; after sudo nano /etc/passwd (closed without saving) and ls /tmp, ausearch -k passwd_changes and -k program_execution return records with user, time and executable; aureport shows counts.

W4-F02 — No working log management

Finding ID	W4-F02
Date / time observed	30 Sep 2026, 12:32–12:41; 13:53 (Lynis scan); 21:49
Evidence source	journalctl; grep on auth.log and syslog; /var/log listing; Lynis LOGG-2130 and LOGG-2138
Evidence summary	journalctl: “No journal files were found”; auth.log and syslog do not exist; wtmp, btmp, lastlog and faillog are 0 bytes; Lynis: no running log daemon, klogd not running. (Figures 12, 13, 14, 2, 24, 21)
Security significance	Failed logins, sudo use, service errors and kernel messages cannot be reviewed or forwarded, so investigation and privileged-access review are not possible.
Control objective	Affected control: C2 (log management and authentication monitoring). Authentication, privilege and system events are logged, retained and reviewable, and available to central monitoring.
Control owner	Security Operations Manager; Platform Engineering
Risk / priority	[object Object]
Recommended action	Install and enable rsyslog or persistent journald with auth and kernel facilities; check login accounting; set up rotation and forwarding.
Retest evidence	Auth and syslog sources populated; a harmless sudo event and a failed-login test appear locally and in the SIEM; Lynis LOGG-2130 and LOGG-2138 cleared.

W4-F03 — Patch backlog and vulnerable packages

Finding ID	W4-F03
Date / time observed	29 Sep 2026, 13:17; 30 Sep 2026, 13:44; Lynis scan about 13:53
Evidence source	apt update output on two days; Lynis PKGS-7392 (warning) and PKGS-7420 (suggestion)
Evidence summary	“91 packages can be upgraded” on both 29 Sep and 30 Sep; Lynis warns that one or more vulnerable packages were found. (Figures 5, 15, 21, 22)
Security significance	Pending updates may include fixes for known vulnerabilities, and the unchanged count suggests nothing was patched in that period.
Control objective	Affected control: C3 (patch and vulnerability management). Security updates are applied within the approved timeframe and verified.
Control owner	Infrastructure / Patch Management Lead
Risk / priority	[object Object]
Recommended action	Review apt list --upgradable; apply security updates in a change window; enable unattended security upgrades.
Retest evidence	apt update shows no pending security updates; PKGS-7392 is absent from a new lynis audit system.

W4-F04 — Hardening gaps, malware protection and file integrity

Finding ID	W4-F04
Date / time observed	30 Sep 2026, scan about 13:53; reviewed 14:14–14:20
Evidence source	Lynis hardening index, kernel hardening, hardening checks and suggestions
Evidence summary	Hardening index 62; 16 sysctl keys DIFFERENT; malware scanner NOT FOUND; no file-integrity tool (FINT-4350); no fail2ban (DEB-0880); password-policy suggestions (AUTH-9230, 9262, 9286, 9328). (Figures 19, 25, 26, 27, 28, 30, 31)
Security significance	A weaker baseline and no malware or integrity monitoring make it more likely that a compromise or tampering goes unnoticed.
Control objective	Affected controls: C4 (hardening baseline) and C6 (malware protection and file integrity). Systems meet the approved security baseline and unauthorised change is detected.
Control owner	Platform Engineering; Endpoint Security Lead; IAM Manager
Risk / priority	[object Object]
Recommended action	Apply the prioritised suggestions under change control; deploy an approved malware scanner and file-integrity tool; put in place password policy and brute-force throttling.
Retest evidence	Related Lynis IDs cleared; hardening index at or above target in the next quarterly run.

W4-F05 — Privileged access cannot be independently reviewed

Finding ID	W4-F05
Date / time observed	30 Sep 2026, 21:49
Evidence source	id output; absence of auth.log
Evidence summary	My account (uid 1000) is in the sudo and adm groups and I used sudo throughout the lab; no authentication log exists to verify that use. (Figures 2, 13)
Security significance	Root-level and log-read rights exist without an independent review trail.
Control objective	Affected control: C5 (privileged-access oversight). Privileged access is authorised and recertified, and its use can be reviewed.
Control owner	IAM Manager; System Owner

Risk / priority	[object Object]
Recommended action	Document the authorisation for the membership; enable sudo logging; schedule quarterly recertification.
Retest evidence	Approved access record and a sample of sudo events in logs or the SIEM.

W4-F06 — Single DNS resolver

Finding ID	W4-F06
Date / time observed	30 Sep 2026, scan about 13:53
Evidence source	Lynis NETW-2705
Evidence summary	Only nameserver 127.0.0.11 is configured; Lynis could not find 2 responsive nameservers. (Figures 23, 21)
Security significance	A resolver failure would hold up updates and any log forwarding.
Control objective	Affected control: C7 (network service resilience). Name resolution remains available to the host.
Control owner	Network Operations
Risk / priority	[object Object]
Recommended action	Add a backup nameserver by the approved method.
Retest evidence	NETW-2705 no longer reported in a new Lynis run.

W4-F07 — Platform process limit and restricted environment

Finding ID	W4-F07
Date / time observed	30 Sep 2026, 21:46–22:04
Evidence source	Terminal output; cat /proc/1/comm
Evidence summary	“sudo: unable to fork”, segmentation faults and “bash: fork: retry” errors; PID 1 is s1eep. (Figures 1, 3, 4)
Security significance	Resource limits interrupted my commands and could just as easily interrupt security tools. The restricted platform also explains why auditd and journald could not run.
Control objective	Affected controls: C1 and C2 (platform dependency). The platform provides enough capacity and services for security controls to operate.
Control owner	Platform owner
Risk / priority	[object Object]
Recommended action	Review process limits and capacity monitoring; confirm which services the platform supports.
Retest evidence	No fork errors in normal use; a capacity dashboard in place.

Appendix B — Evidence Completeness Matrix

This matrix maps each requirement in the lab brief to the evidence I have, including where the environment limited what I could show.

Brief requirement	Status	Where / notes
Document equivalent log sources / methods where my environment differs (Section 3 rules; Activities 1.2, 2.2, 2.3)	Provided	Section 3.1 (one table covering all 8 cases)
Bundle 1: evidence that auditd is running	Provided; shows NOT running	Figures 5, 6, 7; Section 4.1
Bundle 1: loaded custom audit rules	Rules shown; loading blocked	Figure 8; auditctl -l error; Section 4.2
Bundle 1: at least one ausearch result for a configured key	Attempted for all three keys; error output	Figures 10, 11; Section 4.3
Bundle 1: at least one aureport summary	Summary, --failed and --login attempted; error output	Figures 10, 11; Section 4.3
Bundle 1: interpretation of user / action / time	Provided	Section 4.4
Bundle 2: journalctl review	Provided; no journal files	Figure 12; Section 5.1
Bundle 2: authentication / privilege-use evidence	Equivalent sources documented	Figures 13, 2, 24; Section 5.2
Bundle 2: error / warning evidence (syslog or equivalent)	Equivalent sources documented	Figures 14, 21, 4; Section 5.3
Bundle 2: event table	Provided	Section 5.4 (9 rows)
Bundle 3: Lynis version and execution	Provided	Figures 16, 17, 18
Bundle 3: hardening index / summary	Provided (62)	Figure 19
Bundle 3: at least three warnings or suggestions	Provided (3 warnings, 43 suggestions)	Figures 21, 27, 31
Bundle 3: findings with risk, evidence, owner, remediation, priority	Provided	Section 6.5 and Appendix A
Bundle 3: optional retest (before and after)	Not performed (optional)	Section 6.6
Bundle 4: control table (5+ rows; 3+ evidence-based)	Provided (8 rows; 7 evidence-based)	Section 7
Bundle 4: escalation questions	Answered	Section 7.1
Bundle 5: SIEM and automation mapping	Provided	Section 8
Bundle 6: final audit report structure	Provided	This document
Optional: failed local authentication attempt (Activity 1.3)	Not performed (optional)	Optional activity; not carried out

Appendix C — Supporting Screenshot and Command-Output Evidence (Figure Index)

The supporting screenshots and command output (Figures 1–31) are embedded in the section that discusses each one, so the evidence sits beside the analysis. This index lists every figure, with its section and capture time, so that any item can be found quickly.

Figure	Title	Section	Captured (WAT)
1	Operating system, kernel and sudo check	2 Scope	30 Sep 2026, 21:46
2	Identity, PID 1 and /var/log inventory	2 Scope	30 Sep 2026, 21:49
3	System clock, identity, PID 1 and resource errors	3 Methodology	30 Sep 2026, 22:01
4	Shell start-up under process-limit errors	3 Methodology	30 Sep 2026, 22:04
5	auditd status check, package refresh and install attempt (Activity 1.1)	4.1	29 Sep 2026, 13:17
6	auditd package and service verification	4.1	29 Sep 2026, 13:28
7	auditd process check and audit directory inspection	4.1	30 Sep 2026, 11:50
8	Custom audit rules, restart attempt and loaded-rule query (Activity 1.2)	4.2	29 Sep 2026, 13:41
9	Benign event generation (Activity 1.3)	4.3	30 Sep 2026, 11:24
10	aureport and ausearch attempts (Activity 1.3)	4.3	30 Sep 2026, 11:26
11	ausearch for all three keys and aureport summaries (Activity 1.3)	4.3	29 Sep 2026, 09:58
12	journalctl review (Activity 2.1)	5.1	30 Sep 2026, 12:32
13	Authentication and privilege-use log review (Activity 2.2)	5.2	30 Sep 2026, 12:39
14	General system log review (Activity 2.3)	5.3	30 Sep 2026, 12:41
15	Package list refresh before Lynis installation	6.1	30 Sep 2026, 13:44
16	Lynis installation (Activity 3.1)	6.1	30 Sep 2026, 13:45
17	Lynis version	6.1	30 Sep 2026, 13:52
18	Lynis audit execution and environment detection (Activity 3.2)	6.1	30 Sep 2026, 13:54
19	Lynis — scan details and hardening index	6.1	30 Sep 2026, 14:19
20	Lynis — end of scan and exceptions notice	6.1	30 Sep 2026, 14:20
21	Lynis — Warnings (3)	6.2	30 Sep 2026, 14:16
22	Lynis — ports and packages section	6.3	30 Sep 2026, 14:14
23	Lynis — networking and firewall sections	6.3	30 Sep 2026, 14:14
24	Lynis — logging, insecure services and banners	6.3	30 Sep 2026, 14:14
25	Lynis — kernel hardening (sysctl comparison, part 1)	6.3	30 Sep 2026, 14:15
26	Lynis — kernel hardening (part 2), hardening checks and start of results	6.3	30 Sep 2026, 14:16
27	Lynis — Suggestions (43), part 1	6.4	30 Sep 2026, 14:17
28	Lynis — Suggestions, part 2	6.4	30 Sep 2026, 14:17
29	Lynis — Suggestions, part 3	6.4	30 Sep 2026, 14:18
30	Lynis — Suggestions, part 4	6.4	30 Sep 2026, 14:18
31	Lynis — Suggestions, part 5 and follow-up	6.4	30 Sep 2026, 14:19